

Review

A systematic analysis of failures in protecting personal health data: A scoping review

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ARTICLE INFO

Keywords:

Data privacy
Cybersecurity
Personal health data
Data breach
Data protection
Digital health

ABSTRACT

Personal health data breaches pose significant challenges to healthcare providers and clients. This study systematically analyzes 5470 records and reviews 120 articles on this theoretically and practically important topic. It summarizes the existing literature and develops an integrative model with eleven propositions explaining the multifaceted nature of health data breaches, their facilitators, and their impacts. We report on the gaps in the current literature and discuss six promising avenues of future research, including specific suggestions for multi-level analysis, use of novel methods, contributions to information systems theory, stakeholder analysis, under-explored themes, and boundary-breaching opportunities. Beyond these findings, our study offers implications for key stakeholders in healthcare settings. This study equips practitioners and researchers with a valuable model for evidence-based data breach risk management and offers guidance for future investigations, enhancing our collective understanding of personal health data breaches within healthcare.

1. Introduction

Existing research emphasizes the beneficial effects of using digital health systems on various aspects of performance and care quality (Chandwani, De, & Dwivedi, 2018; Kohli & Tan, 2016; Stephanie & Sharma, 2020; Yeung, 2019). Simultaneously, the process of converting personal health records into a digital format has introduced new information risks, particularly in terms of privacy and data security (Harvey & Harvey, 2014; Kordzadeh, Warren, & Seifi, 2016; Kwon & Johnson, 2018). While healthcare providers are showing gradual awareness of cyber risks, the abundance of reports highlighting failures in data protection raises concerns regarding the current practices of the preservation of health information privacy in the digital age. For example, a healthcare provider experienced a significant data breach impacting 8.9 million individuals, including personal and protected health information, with a ransom demand of \$10 million (Alder, 2023). This event was one among several high-profile instances of healthcare data breaches. Notable examples include the notorious theft of 78.8 million patient records from Anthem Blue Cross, a health insurance plan provider based in the United States (US) (Lord, 2018), and the illicit sale of Australian

Medicare card information on a darknet auction site (Elton-Pym, 2017).

In this paper, we embrace the concept of a personal health data breach as “an incident that results in the confirmed disclosure – not just potential exposure – of [personal health] data to an unauthorized party” (Verizon, 2020, p. 4). Personal health data breaches are particularly consequential and can cause serious harm to individuals. Not surprisingly, when examined from the perspective of a hacker, the worth of an individual’s medical record exceeds that of their credit card details by a considerable margin (Humer & Finkle, 2014; Maruca, 2015). While credit card information can make cash easy for a hacker, the impacts of health data breaches are more considerable. For example, investigative reports found that on the darknet, an individual’s medical records could be worth \$100 or even \$1000, while a credit card is worth \$1 or less (Yao, 2017; Zandona & Thompson, 2017). This personal information typically encompasses a considerable volume of confidential information, comprising complete names, contact details, insurance particulars, diagnoses, prescribed medications, and medical procedures. Unlike credit card numbers, this data does not lend itself to easy disposal. More often than not, its relevance persists for extensive duration, occasionally even indefinitely. Cybercriminals possess the ability to exploit

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compromised health data in innovative manners, thereby deriving monetary benefits (Maruca, 2015). Consequently, in the event of a compromise in personal health records, there exists the potential for various deceptive activities, including identity theft, insurance fraud, the fabrication of counterfeit prescriptions, and related offenses.

Emerging evidence posits that healthcare organizations exhibit a higher vulnerability to data breaches in contrast to their counterparts in other sectors (Gordon, Fairhall, & Landman, 2017). This vulnerability arises when multiple actors within healthcare settings have access to personal health data for patient care, increasing the chances of human errors leading to privacy breaches. Inadequate staffing and investments for IT security, as well as inadequate technology investment, further contribute to the heightened risk faced by healthcare organizations. In addition to the potential harm to affected individuals, data breaches result in severe financial and organizational consequences, including regulatory penalties, and damage to the organization's reputation (Goode, Hoehle, Venkatesh, & Brown, 2017; McLeod & Dolezel, 2018).

There is an increasing global recognition of the importance of protecting privacy and personal information (Esmaeilzadeh, 2020; Labadie & Legner, 2023; Miltgen & Smith, 2015; Zuboff, 2015). This growing awareness is reflected in the introduction of new privacy regulations such as the EU's General Data Protection Regulation (GDPR), California's Consumer Privacy Act (CCPA), and China's Personal Information Protection Law (PIPL). Prominent incidents like the Facebook – Cambridge Analytica scandal and the SolarWinds hack have further intensified the global focus on data privacy and security.

Given the importance and implications of healthcare data breaches, alongside the ongoing trajectory toward digitalization and the emergence of data-driven institutions, we believe it is imperative to review occurrences of personal health data breaches in the present era. Existing literature fails to provide an appropriate understanding of health data breaches. The occurrence of data breaches, facilitators, and outcomes in information systems (IS) research and healthcare practice remains unclear. Our review addresses an incomplete view on data breaches, particularly with a clear focus on the healthcare context (see Section 2). As we argue in the following section, the critical importance and idiosyncrasies of data privacy in healthcare organizations call for context-specific investigations (Burton-Jones & Volkoff, 2017; Hassandoust, Akhlaghpour, & Johnston, 2020). Also, scholars have repeatedly raised concerns about the relevance of IS research for information technology (IT) practitioners (Moeini, Rahrovani, & Chan, 2019). Information security research should inform data security decisions and motivate the effective protection of digital assets (Vedadi, Warkentin, & Dennis, 2021). Contextual investigation can help IS research to be relevant and useful (Davison & Martinsons, 2016). In this study, we re-examine data breach phenomena with a novel context-specific perspective, focusing on healthcare settings and actors.

To gain a comprehensive understanding of the circumstances in which healthcare organizations are vulnerable to cyber threats and the potential negative consequences, we aim to synthesize the mechanisms of data breaches in the literature. Our study is based on a scoping review that delves into the research question: "What insights can be drawn from the existing literature concerning the process of personal health data breaches and their consequent effects?" We also aimed to integrate theoretical and empirical perspectives on failures in data protection and show how the integration of these perspectives can enrich the understanding of data protection phenomena and further theorization.

Our study contributes to better understanding the occurrence of different health data breach types and their facilitators and outcomes in a way that informs policy and practices. Specifically, we delve into the mechanisms underlying data protection failures unique to the healthcare setting, enabling healthcare managers to proactively address and mitigate the risks associated with different breach types. Thus, managers can meaningfully apply the insights gained from our findings. The results of our research can also provide valuable insights to healthcare organizations, policymakers, and cybersecurity professionals involved

in the development and implementation of digital healthcare systems. These insights can assist them in improving the security and privacy of patients' health information. Furthermore, this study reveals existing gaps in knowledge and identifies potential areas for future investigation.

The remainder of this paper is organized as follows. We begin by outlining the review methodology, which encompasses the article selection and coding process. The subsequent section presents the findings, wherein we provide concise summaries of the various types of data breaches, along with the factors that facilitate them and the impacts of data breaches. These findings are categorized into multiple sub-sections. Moving forward, as an integrative approach, we illustrate a path towards theorizing data protection. This path includes an analysis of the theoretical paths in a hybrid (process and variance) model, and stakeholder analysis. Finally, we discuss fruitful future research directions, and conclude.

2. Addressing an incomplete view on health data breaches by a contextual investigation

A general investigation of a phenomenon without contextualization offers limited insights and can hardly inform practice (Davison & Martinsons, 2016). Excluding the context in studying organizational practices leads to an incomplete conclusion and underplays the research implications (Jarzabkowski, Kaplan, Seidl, & Whittington, 2016). In our review, in line with multiple calls for a context-specific investigation (Avgerou, 2019; Hong, Chan, Thong, Chasalow, & Dhillon, 2014; Pentland, Recker, Wolf, & Wyner, 2020), we argue that healthcare, as an extreme context (Hällgren, Rouleau, & De Rond, 2018), requires context-specific attention in studying the data breach phenomena.

As noted by Burton-Jones and Gallivan (2007) context is associated with the way that patterns of information system use emerge in organizations. Users, systems, and tasks as key elements of system use constitute effective use of IT (Burton-Jones & Grange, 2013). These elements are context-specific and influence effective use actualization (Burton-Jones & Volkoff, 2017). For instance, in the healthcare context, the ways health professionals (users) use features of electronic medical records (EMR) systems to perform clinical tasks and protect patient data are specific to the context. Thus, effective data protection practices and cybersecurity risk management in healthcare requires contextualization. Further, in the healthcare context, multiple actors (i.e., clinicians, residents, nurses, managers, IT staff, and administrative staff) are involved in the implementation and use of digital health (Burton-Jones & Volkoff, 2017; Negoita, Rahrovani, Lapointe, & Pinsonneault, 2022). The challenges and complexities these actors experience are unique (Wall, Lowry, & Barlow, 2015). Moreover, the use of health information systems is heavily regulated when compared to other forms of information systems (Keith et al., 2022). Variations, tensions, and conflicts exist in this context due to different clinical groups and positions of power between physicians and nurses, as well as professional and administrative logic (e.g., patient care, use of funds, governance structure) (Burton-Jones et al., 2020; Lapointe & Rivard, 2005; Skyvell Nilsson, Törner, & Pousette, 2018). Similarly, as indicated by Vaast (2007), based on a social representations perspective, there are different logics among healthcare communities in terms of understanding information security and data protection practices. Vaast highlighted that the context of practice should be taken into account to understand information security issues in healthcare. Inadequate security of digital assets and breaches of customer data are consequential for every organization. However, healthcare is an extreme context and needs special considerations in data protection (Culbertson, 2021; Osterman Research, 2020). Healthcare is among the top industries affected by cybersecurity incidents (NNT, 2021; Zandt, 2022). Data breaches are more costly for healthcare organizations and can influence patient life, according to an IBM investigation on the cost of a data breach (IBM Corporation, 2021) and the Ponemon Institute report on the healthcare security ecosystem (Ponemon Institute, 2021). From a consumer perspective, health data

privacy is a highly consequential and controversial subject (Agarwal, Gao, DesRoches, & Jha, 2010).

As personal health data is extremely sensitive, patients are concerned about how healthcare organizations and health professionals use and process their data. Unauthorized disclosure of health data such as mental health details can affect patient social acceptance and trust (Coventry & Branley, 2018; Kohli & Tan, 2016). Hence, given the need for context-specific investigations in healthcare (Davidson, Baird, & Prince, 2018; Mindel & Mathiassen, 2015; Rasmussen & Sahay, 2021) and the importance of understanding data breach phenomena (Angst, Block, D'arcy, & Kelley, 2017; IBM Corporation, 2021; Khan, Kim, Mathiassen, & Moore, 2021), we believe reviewing health data breaches is timely and appropriate for offering contextually relevant insights.

To set the foundation for our context-specific investigation into the state of the existing knowledge on health data breaches, we start by establishing the definitions of relevant terms in our study. A healthcare data breach refers to a privacy or cybersecurity incident that leads to the unauthorized exposure and use of health data. The definition we adopted is based on the data breach investigation report by Verizon (2020). In the context of health data breaches, the definition states that a breach occurs when there is a confirmed disclosure, rather than just the potential exposure, of personal health data to an unauthorized party. We also use the term “health data breach” to encompass various types of breaches, including unauthorized access, hacking, or IT incidents. Data breach facilitators are conditions and failures that contribute to health data breaches, such as a lack of cybersecurity awareness. The impact of a health data breach refers to the adverse effects and influences that occur as a result of a data breach incident. The breach affects numerous stakeholders, including healthcare customers or patients, health professionals, healthcare managers, the IT department, and organizations. These impacts include concerns regarding health privacy, hindered health information sharing, and compromised care quality.

As part of our context-specific investigation, we re-examine the current state of reviews on data breaches. To do this meta-review (review of reviews), we evaluate past review papers based on multiple criteria including their elucidation of search strategy, study selection, list of included studies, data breach types, data breach facilitators, data breach impacts, data breach theoretical model, and research agenda. This re-examination helps to illustrate how past research has approached data breach phenomena in healthcare (see Table 1). Criteria such as transparency and research agenda are vital for the accumulation of knowledge in a research stream and for research impact (Paré, Tate, Johnstone, & Kitsiou, 2016; Wagner et al., 2021). In our evaluation of past reviews, we identified these criteria were inadequately fulfilled. These reviews provide a useful but narrow path towards an understanding of health data breaches. Based on our synthesis of the past reviews, the explanations on the process of health data breaches are incomplete, especially with regard to the impact of data breaches. We argue a review is needed that addresses the above shortcomings, in order to provide researchers and data protection practitioners with more informed conclusions and interpretations on data breaches. We seek to address critical shortcomings in the past reviews and extend them along several dimensions, including a focus on the data breach process, impacts, demonstration of methodological transparency, and research agenda.

3. Review methods

The study conducted a scoping review, which is appropriate when there is a need to explore a broad topic using various research methods and address wide-ranging research inquiries (Levac, Colquhoun, & O'Brien, 2010). It also helps to identify research gaps in extant literature to inform future research and practice (Paré, Trudel, Jaana, & Kitsiou, 2015; Schryen, Wagner, Benlian, & Paré, 2020; Templier & Paré, 2015a). The scoping review as a method has been well established in the healthcare domain. It is also applied extensively in other disciplines,

including management, implementation science, psychology, computer science, informatics, and IS (Archer, Fevrier-Thomas, Lokker, McKibbon, & Straus, 2011; Chung et al., 2020; Courtin & Knapp, 2017; Davis, Drey, & Gould, 2009; Dowdeswell, Sinha, & MacDonell, 2020; Gagliardi, Berta, Kothari, Boyko, & Urquhart, 2015; Gutt, Neumann, Zimmermann, Kundisch, & Chen, 2019; Li, Marier-Bienvenue, Perron-Brault, Wang, & Paré, 2018; Szasz, Palinkas, Palfi, Szollosi, & Aczel, 2018). To conduct a scoping review, we adhered to the instructions provided by Arksey and O'Malley (2005). The process involved five distinct stages: formulating the research question, identifying relevant studies, selecting appropriate studies, organizing the collected data, and summarizing and presenting the findings. For our review, we conducted a systematic search with particular focus on evidence-based literature. We sought to provide a comprehensive image of failure in data protection in healthcare cybersecurity to inform practices. As this review is exploratory (Paré et al., 2015), we also provided a road map for future research.

To identify relevant research studies, a thorough search was conducted in two prominent academic databases: PubMed and Scopus. The search was centered on the comprehensive term “data breach”. We conducted the search in these databases, utilizing specific search terms (including data breach*, security breach*, breach notification, information breach*, privacy breach*, cybersecurity breach*, privacy invasion, unauthorized access, disclosure of information, information disclosure*, disclosing information, disclosing health data, disclosing data, information exposure, data exposure, security leak*, data leak*, information leak*, cybersecurity leak*, leaked data, leaked security and leaked information). Our review illustrates literature on health data breaches from 2009 to 2022 to show contemporary perspectives on security and privacy events. The timeline for the literature search represents the digitalization era (Teubner & Stockhinger, 2020), which is consistent with the focus of our review. Particularly in the healthcare domain, the uptake in the adoption of electronic health (or medical) records (EHR/EMR) happened in this time frame. For example, in the US, the Health Information Technology for Economic and Clinical Health (HITECH) Act, which introduced incentives and penalties for EHR/EMR adoption, was enacted in 2009. A surge in the diffusion and use of these systems¹ inevitably led to heightened and new health data breach risks. Also, as ways of data breaches and the use of digital health technologies are evolving, focusing on the current literature provides more relevant and useful insights into data protection practices.

The identified records retrieved from the database were subsequently transferred to the citation manager, EndNote, for further processing. Consistent with the guidelines for conducting a scoping review, a set of inclusion and exclusion criteria was established prior to the examination of the records and full texts of the articles. Details of the criteria that constitute the study selection process are presented in Table 2.

Data extraction was facilitated by using a template that covers authors (year), journal, country, methods, systems, context, theory, level of analysis, data breach types, data breach facilitators, and impacts. Summarization and sense-making of extracted data were performed via thematic analysis (Templier & Paré, 2015b). Using this method for research synthesis has been applied in IS and medical informatics (Meyer & Paré, 2015; Recker, Indulska, Green, Burton-Jones, & Weber, 2019). This analysis was conducted to identify the scope of literature and report relevant findings in terms of themes (types, facilitators, and impacts of data breaches) that can be identified from the extant studies. Our approach to thematic analysis is also consistent with a retroductive approach (Mueller & Urbach, 2017). The themes emerged from the included empirical studies and then were enriched and refined by cycling between data and IS literature for providing qualitative rigor

¹ For data on the diffusion of EHR/EMR systems after 2009, please see <https://dashboard.healthit.gov/quickstats/quickstats.php>

Table 1

Past review studies on data breaches and shortcomings.

Article	Search strategy			Study selection		List of included studies	Data breach types	Data breach facilitators	Data breach impacts	Data breach theoretical model	Research agenda
	Keywords	Database	Date of coverage	Inclusion and exclusion criteria	Process of screening and selection						
Coventry and Branley (2018)	✓	PubMed	2014–2018	✓	Not Clear	NA	Limited to cybersecurity	NA	Cited a few impacts without in-depth explanations	NA	NA
Kruse, Frederick, Jacobson, and Monticone (2017)	✓	PubMed, CINAHL, ProQuest	2006–2016	✓✓	✓	31 articles	Limited to hacking/ IT incident (e.g., cyberattacks, ransomware)	NA	NA	NA	NA
Kamoun and Nicho (2014)	NA	NA	NA	NA	NA	NA	✓	✓	NA	Swiss Cheese Model	NA
Chernyshev, Zeadally, and Baig (2019)	NA	NA	NA	NA	NA	NA	✓	NA	NA	Conceptual architecture for digital forensic characteristics	NA
Zandona and Thompson (2017)	NA	NA	NA	NA	NA	NA	✓	NA	NA	Framework for data protection	NA
Wu, Torous, Hardaway, and Gutheil (2017)	NA	NA	NA	NA	NA	NA	✓	NA	Focusing on trust	NA	NA
Yeng, Yang, and Sneekenes (2019)	NA	IEEE Xplore, Google Scholar, Science Direct	NA	✓	NA	NA	NA	✓	NA	NA	NA

(Gioia, Corley, & Hamilton, 2013). This approach is evident in themes (e.g., noncompliance) that were constructed from our synthesis.

Our approach to thematic analysis aligns with the methodology recommended by Gioia et al. (2013) for achieving qualitative rigor in inductive research. The steps we followed can be elucidated as follows:

1st order concepts: We initiated our analysis by identifying relevant concepts related to data breaches from the included studies. This initial coding process, known as 1st order coding with open coding, involved extracting concepts that emerged from the literature.

2nd order themes: In the next stage, we delved deeper into these concepts to assess whether they could effectively describe and explain an underlying theme and phenomena of health data breaches that we were observing across the included studies. During this phase, we began to group these concepts into second-order themes.

Aggregating themes into dimensions: These second-order themes were then further aggregated into dimensions. The second-order themes contributed to the identification of the overarching themes related to data breaches. These aggregate dimensions encompassed the types, facilitators, and impacts of data breaches. By following this approach, we were able to gain an understanding of the health data breach landscape and provide a structured synthesis of the findings from the reviewed studies.

In the appendix of our paper, we have included information that represents our coding process, starting from the data extraction table to the creation of several tables that present the findings. Specifically, we extracted the following information from each full-text article: author/s, country, setting, system, method, and level of analysis. This information helped us understand the context and characteristics of the reviewed articles. Furthermore, we categorized the data breach stakeholders based on their roles, such as IT/security department, healthcare

personnel/staff, health manager, and business associates. We also coded the different types of data breaches themes (e.g., hacking or IT incidents, unauthorized access/disclosure, loss, theft, improper disposal), facilitators of data breaches (e.g., noncompliance, lack of awareness, third-party involvement, inappropriate collective responses, organizational characteristics), and the impacts of data breaches (e.g., privacy concerns, system use, trust, health information sharing, care quality, trust repair costs). We conducted inductive coding to categorize stakeholders affected by data breaches, identify the factors that enable data breaches, and assess the consequences of such breaches. In the coding process, the first authors primarily conducted the coding of the articles. The second and third authors reviewed the coding process to ensure accuracy and consistency. In cases of disagreement, the three of us discussed until consensus was reached. This multi-author approach helped minimize potential bias and enhance the reliability of the coding.

Furthermore, as a knowledge-development activity in this scoping review study, we went beyond summarizing the extant literature on health data breaches. We developed a model of health data breaches. By targeting a forward-oriented approach (Schryen et al., 2020), we also sought to draw up a research agenda for the further theorization of health data breaches. The process of our review was guided by recommendations for systematic and transparent reviews (Paré et al., 2016).

4. Findings

4.1. Study selection

Fig. 1 provides an overview of the selection process and outcomes at each stage, specifically the identification, screening, eligibility, and inclusion of articles in the study. From the identification step, 5470 article

Table 2
Inclusion and exclusion criteria.

	Inclusion criteria	Exclusion criteria
Language	English	Non-English
Full-text availability	Full text	Full text not available
Publication dates	2009–2022/08/03	Before 2009 and after 2022/08/03
Source type	Journal (Empirical (original /research articles) and non-empirical articles (review, conceptual), peer reviewed	Editorial, comment and response, letter, perspective, abstract only, conference, and symposium proceedings paper, book section
Subject area and method	Health data breaches (e.g., unauthorized access, use, data collection, and disclosure of health information)	Irrelevant to data breaches, the sole emphasis lies on: encryption, protocols, algorithms or mathematical models, simulation, law/rule and ethics, scenarios, learning models, implementation, design or architecture, development of computer programs, modeling language (e.g. UML)
Type of system	Digital health, systems containing personal health data and related to the study setting	Systems lacking the capability for personal data processing
Data	Electronic/Digital	Non-electronic, e.g., paper data
Participants	Individuals (actors committed unauthorized access, hackers, healthcare staff, healthcare professionals, medical students, physician assistant (PA), and allied health professionals)) or organization (see study setting)	Users not contributing to health data processing
Study setting	Health services and organizations providing the services	Not health data setting
The findings of the study	Concerning the processes/ and impacts of data breaches	No relevant data breach findings

records were retrieved. We removed 1177 duplicated records. In the next step, we review 4293 records, following an in-depth evaluation of 371 full-text articles. Finally, 120 articles met the inclusion criteria and were included in the final analysis. Following that, an extraction of detailed data from the included articles was carried out (see Appendix, Table A1).

4.2. Study characteristics

Numerous studies on data breaches have recently been conducted, as depicted in Table 3. Some 42% of the articles included in these studies were published between 2017 and 2019. The results indicate that a significant portion of the data breach literature originates from North America, particularly the US, while academic insights from other regions are limited. The studies encompass a wide range of settings, with a notable emphasis on hospitals. The table illustrates the examination of diverse systems, wherein health information systems (HIS) in general, and mobile health and EHR/EMR systems specifically, are the most commonly investigated ones. An interesting discovery from Table 3

reveals that the majority of studies utilized quantitative methods and focused on the individual level of analysis. Notably, only one study was undertaken in the realm of group-level analysis. Finally, only a limited number of studies have specifically applied or been informed by a theory. The examination shows that 29 publications, conducted at various analytical stages, drew inspiration from established theories.

In short, the majority of included articles were published between 2017 and 2019, conducted quantitatively, and analyzed at the individual level in the US-dominated context.

4.3. Thematic analysis

For the thematic analysis, we sought to construct a high-level framework as a template to organize emerging themes. Based on a hybrid view, integration of variance, and process thinking recommendations (Burton-Jones, McLean, & Monod, 2015; de Guinea & Webster, 2017; Langley, 2007; Maitlis & Lawrence, 2007), for a systematic examination of the health data breaches, we developed a model (see Fig. 2). This model distinguishes between facilitators (the conditions that clear the paths for potential data breaches), data breach types, and impacts (an effect or influence that a data breach incident has on individuals and organizations/businesses). In developing our model and identifying antecedents, facilitators, and impacts, we were inspired by recent methodological developments in the IS field (Burton-Jones et al., 2015; de Guinea & Webster, 2017) and the notion of hybrid models. In our model, facilitators and data breach types (enablers) denote a process approach (see, for example, Maitlis and Lawrence, 2007), while studying impacts (consequences) pertains to variance approaches. As stated by de Guinea and Webster (2017, p. 159), a hybrid model is an ideal candidate for “investigating events that trigger certain changes in states or outcomes”.

An important note should be made that the model in Fig. 2 is the outcome of our synthesis. In a purist rendition of qualitative research, this model should appear after we explain our data and emerging themes. However, following the recommendation of Suddaby (2006), we present the final model first so that the reader notices what the major theoretical dimensions and contributions are likely to be. We then proceed with explaining our findings that led to the development of this model. The themes and concepts of our model and their definitions are provided in Table 4. We provide more details on the coding process and examples of studies in Table A2 of the Appendix.

To establish the connection between the thematic analysis and our model, we have structured our explanation as follows. We begin by introducing the focal phenomena of our model, health data breaches and

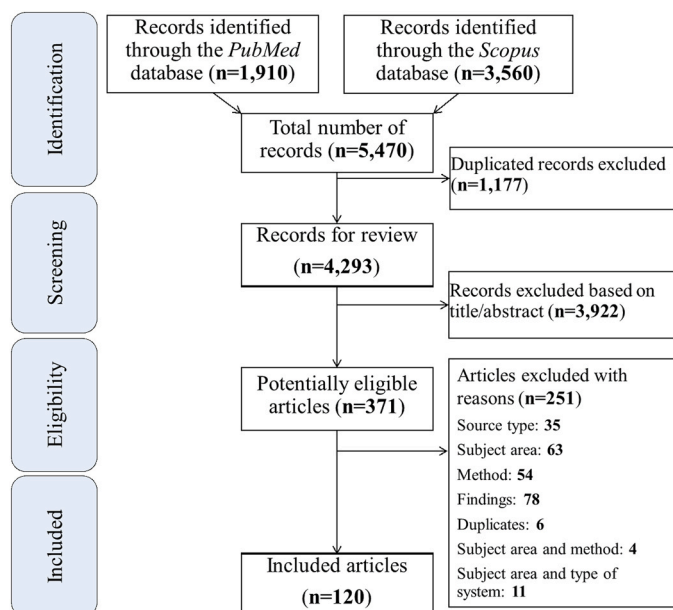


Fig. 1. Study selection process and results.

Table 3
Overview of the studies characteristics.

Variable	Number of studies	Percentage
Year of publication	(13, 13, 12) 38	32
2020–2022	(14, 18, 18) 50	42
2017–2019	(3, 5, 5, 8) 21	17
2013–2016	(3, 1, 3, 4) 11	9
2009–2012		
Region	62 (55,7)	52
North America (the US and Canada)	1	1
Brazil	11	9
Europe	16	13
Asia	5	4
Australia	2	2
Africa	3	2
Multiple countries	20	17
Not applicable		
Systems	50	42
Health Information System (HIS)	17	14
Mobile health (i.e., mobile phones and apps)	15	13
EMR/EMR	8	7
Multiple systems	5	4
Online platform	4	3
Social media	9	7
Other systems (Contact tracing system, Desktop computers, PACS, Portable devices, Privacy-enhancing software, Robotic System, Telehealth, Ultrasound, Email)	12	10
Not applicable		
Setting	10	8
Education	29	24
Hospital	30	25
Multiple setting	12	10
Others (e.g., health research organizations)	3	2
Public health	3	2
Medical center	2	2
Mental health	2	2
Emergency Medicine	1	1
Family Medicine	1	1
Genetic Test Services	1	1
Oncology practices	1	1
Pharmacy	1	1
Physicians' offices	1	1
Psychiatry Outpatients	1	1
Aged Care	1	1
Cardiology	1	1
Dermatology	20	16
Not applicable/unknown		
Method	73	61
Quantitative	12	10
Qualitative	21	17
Review	8	7
Mixed	6	5
Others		
Level of analysis	57	47
Individual	1	1
Group	37	31
Organizational	3	2
IT artifact (Computational view)	2	2
Policy	20	17
Not applicable		
Nature of study	29	24
Theory-driven	70	58
Data-driven	21	18
Literature-driven (without theorizing)		

the associated five types. Next, we present five propositions that link data breach facilitators (noncompliance, lack of awareness, third-party sources, inappropriate collective responses, organizational characteristics) to health data breaches. Then, through the next six propositions, we explain different impacts of health data breaches on privacy concerns, health information sharing, system use, trust, care quality, and trust repair costs, as identified in our review.

4.4. Data breach types

In our framework, data breach types refer to actions that can directly cause a data breach incident to occur. We categorized the breach types into five second-order themes, namely, unauthorized access/disclosure, hacking/IT incident, improper disposal, loss, and theft. These themes were adopted from the US Department of Health and Human Services Breach's portal and were confirmed by our data.

4.4.1. Unauthorized access/disclosure

In our review, a significant portion of the studies (44%) examined unauthorized access and disclosure of personal health data. These breaches occurred in various forms, including cases where authorized employees or third parties were involved in unauthorized usage (Kierkegaard, 2012). Taking Memorial Hermann Hospital in the US as an illustrative case, an incident unfolded involving a former employee who illicitly accessed the EMRs of over 10,000 customers (Bhuyan, Bailey-DeLeeuw, Wyant, & Chang, 2016). Unauthorized access can also occur when medical and health science students, who have not received adequate privacy training, gain entry to confidential information (Calhoun, Kiel, & Morgan, 2018).

A prior investigation revealed that a significant proportion of residents, fellows, and students relied on their personal mobile phones for the purpose of storing and transferring clinical images (Guo, Phan, Ho, Pawlovich, & Kitson, 2018). In the context of mobile health (mHealth), a study shed light on a noteworthy observation. Despite having received training on Health Insurance Portability and Accountability Act (HIPAA) compliance and being familiar with the regulations pertaining to electronic communication, a significant number of surgeons persist in using their personal smartphones to transmit personal health data via text messages (Drolet, Marwaha, Hyatt, Blazar, & Lifchez, 2017). While recognizing the inherent security risks associated with this communication method, the utilization of personal devices by students to share personal health data can likely be attributed to the user-friendly nature of the method (Prochaska, Bird, Chadaga, & Arora, 2015). This emphasizes the importance of mitigating insider threats in the event of a data breach and understanding unauthorized access patterns within the context of organizational data protection failures.

Although healthcare institutions invest in security infrastructures as a proactive strategy to minimize the likelihood of experiencing cybersecurity failures (Kwon & Johnson, 2014), violations of privacy can still occur due to employees and students sharing access credentials (Hassidim et al., 2017). Baskaran, Davis, Bali, Naguib, and Wickramasinghe (2013) reported that a significant number of participants in their study shared their personal passwords with staff members to gain access to their personal health data. This behavior raises concerns and warrants investigation. An underlying explanation for this inappropriate practice could be the participants' underestimation of the potential consequences of a breach in personal health data (Elger, 2009). Unauthorized disclosure of login credentials can also be made via unsecure connections of mHealth applications (Müthing, Jaschke, & Friedrich, 2017). Thus, unauthorized access/disclosure originates from user behavior and system design as facilitators.

4.4.2. Hacking/IT incident

Hacking and IT incidents caused by technical intrusions constitute a major type of many high-impact data breaches (Almulihi et al., 2022). An analysis of reported data from 2009 to 2017 in the US showed that compared to other data breach types, hacking/IT incidents are not the most frequent trigger. However, in terms of impacts, this trigger type was found to affect the largest number of people (approximately 120 million individuals among cover entities) (Yaraghi & Gopal, 2018). Similarly, a study in hospital settings reported that while the number of hacking/IT incidents ranked fourth among six types of breaches, their serious consequences were the highest, affecting over 4600,000 individuals (Gabriel, Noblin, Rutherford, Walden, & Cortelyou-Ward,

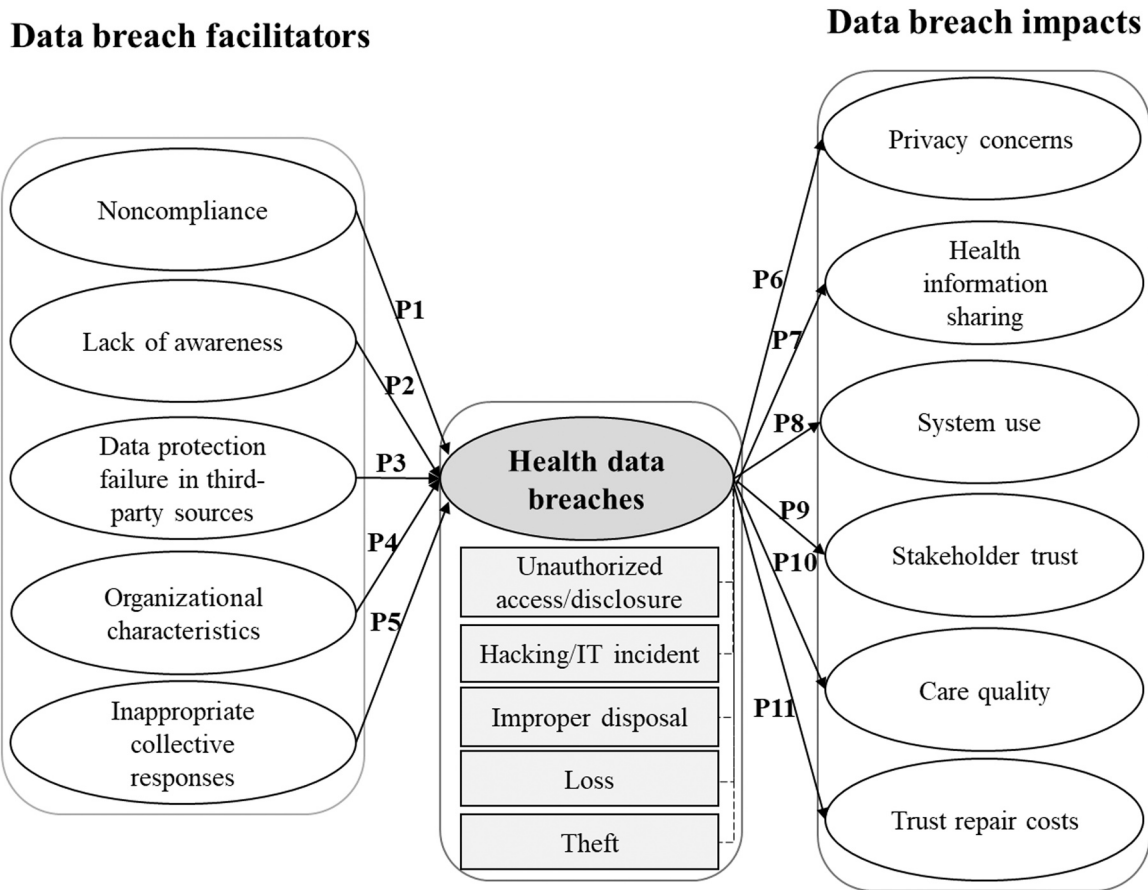


Fig. 2. A model of data breaches based on the scoping review.

2018). Generally, the data from the US Office for Civil Rights (OCR) reveals that hacking incidents hold great significance. It is imperative to employ modeling techniques that can predict the reasons and mechanisms behind these breaches, thereby enhancing the safeguarding of

personal health data.

4.4.3. Improper disposal

Typically, fewer breaches occur by improper disposal. This type was

Table 4
Definitions of components of the health data breaches model.

Aggregate dimensions	Themes	Contextual definition
Type	A <i>health data breach</i> refers to a form of privacy or cybersecurity incident that results in the actual exposure of health data in an unauthorized manner.	
	Unauthorized access/disclosure	Accessing/disclosing health data without permission or in an illegal authority.
	Hacking/IT incident	Accessing health data through technical intrusions by malware or hacking health information systems (e.g., network server, EMR).
	Improper disposal	Inappropriately clearing, disposing, or destroying health data.
	Loss	Losing equipment, including computer desktops, laptops, and other portable devices (e.g., mobile phones), containing personal health data.
Facilitator	Theft	Theft of equipment hosting personal health data (e.g., computer desktops, mobile phones, laptops).
	<i>Data breach facilitators</i> refer to the conditions and failures that clear the paths for potential data breaches.	
	Noncompliance	Failure or refusal to fully comply with health information security policy, rules, and regulations.
	Lack of awareness	Limited knowledge or general understanding of information security policy and health data breach risks/possibilities.
	Data protection failure in third-party sources	Existence of third-party sources (e.g., mobile health (mHealth) apps) that lack data protection by design or have security vulnerabilities for health services.
Impact	Organizational characteristics	Attributes, properties, and resources that are specific to a healthcare organization, such as the size of hospitals, teaching hospitals, and IT budget.
	Inappropriate collective responses	Failing to appropriately respond to data breaches (e.g., hacking) in a collective manner.
	<i>Data breach impacts</i> refer to an adverse effect or influence that a data breach incident has on customers/patients, service providers, and organizations.	
	Privacy concerns	Expressing concerns about improper use, unauthorized access/discourse, error, secondary use, and lack of control over health data processing from data collection to data consumption.
	Health information sharing	Withholding sharing health information for providing or receiving health service delivery.
	System use	Reducing or impeding the use of health information systems (e.g., mHealth, EMR) to perform personal/organizational health task(s) (e.g., by patients or health professionals).
	Stakeholder trust	Damaging the belief and positive expectations that healthcare organizations and their business associates are properly processing and safeguarding health data.
	Care quality	Compromising the quality of healthcare services to patients and health customers.
	Trust repair costs	Incurring costs associated with repairing and enhancing healthcare customers' trust after a data breach, such as advertising, branding, digital forensics, and customer relationship management.

found to be the least common, with minimal individuals affected (Yaraghi & Gopal, 2018). However, there were still instances of this data breach type in our sample, and for effective data protection, organizations must consider the risk of data breaches and the impacts arising from this type of breach.

Electronic health data can be recovered from storage media (e.g., hard disk drives). Medical equipment, including imaging systems that store personal health data, must undergo thorough inspection and employ recovery methods prior to being returned to lease companies or sold to third parties. A comprehensive study involving various methods, such as preliminary disk inspection, deletion techniques, and file recovery, was conducted on ultrasound systems. Surprisingly, it was discovered that personal health data could still be retrieved even if it had been deleted following the systems' prescribed instructions (Moggridge, 2017). Hence, it is evident that there is an urgent need for enhanced and more stringent requirements and standards regarding the appropriate disposal of personal health data.

4.4.4. Loss

This type of data breach refers to the loss of equipment, including computer desktops, laptops, and other portable devices (e.g., mobile phones) containing personal health data. With the increasing use of bring your own device (BYOD) for health service delivery, a data breach such as loss can be a critical challenge for healthcare providers (Wani, Mendoza, & Gray, 2020). There are calls in the extant literature for investigating this data breach type from a socio-technical and socio-ecological viewpoint; that is, interrelations among health information technology, user errors, and environment should be considered in a holistic manner (Johnston, Di Gangi, Howard, & Worrell, 2019; Sittig & Singh, 2011). However, a limited body of literature investigated loss as a type of data breach, and the interpretations were mainly based on secondary data provided by OCR (Koczkodaj, Masiak, Mazurek, Strzalka, & Zabrodskii, 2019a). Wikina (2014) was the first attempt, in the included articles, to study loss as a data breach type. The findings revealed that between 2009 and 2013, theft and loss significantly impacted more individuals compared to hacking or IT incidents. However, in subsequent years the digital transformation in the US healthcare sector reversed this trend, with IT hacking becoming more prevalent (Yaraghi & Gopal, 2018).

4.4.5. Theft

The final type of data breach in our study is due to theft. This data breach type occurs when equipment hosting personal health data is stolen. As highlighted in the OCR reports, theft incidents have occurred across a range of systems, including computers, portable devices, and network servers. Compared to other industries (educational, retail, and government), computer theft is reported as the biggest threat to healthcare settings (Gupta & Sharman, 2012). Often, the very features that enhance the user-friendliness of digital records, such as a centralized database enabling remote access to all health records, inadvertently create opportunities for theft (Arthur Conklin & McLeod, 2010). Theft incidents also could arise from a lack of basic security precautions. For example, in a health research organization, loss of submaster keys (that could provide access to over 60% of a particular office building) caused thefts for around one month (Johnston et al., 2019). In the US context, research published between 2018 and 2019 has revealed that theft is one of the most prevalent types of data breaches. In terms of its impact on individuals, theft ranks second, surpassed only by hacking/IT incidents, and significantly higher than the other three forms of data breaches (Gabriel et al., 2018; Koczkodaj, Mazurek, Strzalka, Wolny-Dominiak, & Woodbury-Smith, 2019b).

4.5. Data breach facilitators

In our thematic analysis, we identified six second-order themes that facilitate the occurrence of the data breach incident. These facilitators –

noncompliance, lack of awareness, data protection failure in third-party sources, organizational characteristics, and inappropriate collective responses – are discussed below.

4.5.1. Noncompliance

One important factor contributing to the violation of privacy and disclosure of personal data is the lack of compliance with information security policies (Gwebu, Wang, & Hu, 2019; Lowry & Moody, 2015; Soomro, Shah, & Ahmed, 2016). According to the American Institute of Certified Public Accountants, it is the second most serious concern in the context of IT management (Warkentin, Johnston, & Shropshire, 2011). Our findings reinforce the view that employees' noncompliance with information security policy results in failure costs for organizations (Dhillon, Talib, Yakimini, & Picoto, 2020; Kwon & Johnson, 2013b).

Past research on compliance intention and behavior has highlighted the role of users' habit formation in compliance with IS policy (Moody, Siponen, & Pahnala, 2018). For instance, Kuo, Chen, Talley, and Huang (2018) conducted a study examining the interplay of motivations (both intrinsic and extrinsic), habit formation, and intention regarding adherence to EMR privacy policy. Their findings revealed that motivational aspects such as self-efficacy and perceived usefulness played a crucial role in fostering habitual behaviors, which subsequently contributed to individuals' sustained commitment to comply with the privacy policy.

Privacy and security rule violations frequently occur when medical and health science students disclose personal health data for non-care-related purposes. Such noncompliance undermines the principles of privacy and security that are crucial in the field of healthcare. For instance, Calhoun et al. (2018) highlighted a concerning incident involving a breach of privacy. In this case, a student attempted to conceal a patient's personal information when submitting an assignment. However, despite the student's efforts, the faculty and staff members were able to identify the patient. This breach exemplifies a failure to comply with established guidelines and safeguards that are in place to protect patient privacy.

In certain healthcare organizations, collaborative challenges arise in protecting patient privacy due to perception gaps among different organizational roles. For instance, it has been observed that users often neglect to log out of their accounts due to the computational inconvenience associated with the login and logout process. On the other hand, IT staff hold the belief that such behavior could compromise patient privacy (Eikey, Murphy, Reddy, & Xu, 2015). Similarly, in an assessment of IT security programs, Hepp, Tarraf, Birney, and Arain (2018) discovered that leaving workstations unlocked was a prevalent form of noncompliant behavior contributing to security breaches. An example of this noncompliant behavior is also reported in a survey of nurses' security practices in Saudi Arabian hospitals (Albarrak, 2012). This kind of failure can stem from poor communication between data controllers and employees, and hamper compliance actions (Desouza & Valverde, 2016). To better understand noncompliant intention and behavior, one should consider the whole system, including communication, leadership and data governance (Barlow, Warkentin, Ormond, & Dennis, 2018; Feng, Zhu, Wang, & Liang, 2019; Renaud, 2012).

Noncompliance can occur due to the ineffective use of information systems. Building on the effective use theory lens (Burton-Jones & Grange, 2013), we define ineffective use as the extent to which individuals and organizations use information systems in a way that fails to achieve the data protection goal. The ineffective use of medical systems can originate from a symbolic (as opposed to substantive) use of data protection features. Failing/refusing to engage in activities (adaptation and training) by employees that would help them become more effective users can also perpetuate ineffective use as a data breach facilitator. For example, employees that are not properly trained to use security protocols may allow vulnerabilities into the systems and increase the risk of a data breach (Angst et al., 2017). In our data extraction, we also found the potential impact of ineffective use of

systems on patient privacy invasions. According to the responses of a sample of residents, ineffective use of technology (e.g., lack of experience in using EMR) facilitates failures in data protection, such as loss of patient records (Meehan, Kawalec, Caldwell, & Putman, 2018). Other paramount concerns to healthcare organizations were raised from the ineffective use of email, one of the main sources of data leakage (Desouza & Valverde, 2016; Gordon et al., 2019). Arain, Tarraf, and Ahmad (2019) found that in a large healthcare organization, the most common stated breaches originated from not knowing how to use the encryption function in communication via email. The adverse impact of ineffective use may not occur or actualize in the short term. However, over time, ineffective use such as symbolic adoption has the potential to hamper IS value. The hidden impacts of this failure will surface in the long term by weakening the relationship between IS security investment and effective data protection; that is, more investment does not lead to better data protection because of predominantly symbolic adoption (Angst et al., 2017). Hence, we put forward the following proposition:

Proposition 1. Healthcare workers' noncompliant behavior can facilitate data breach incidents.

4.5.2. Lack of awareness

The information system literature emphasizes the importance of security awareness as a countermeasure to mitigate IT misuse and promote employees' compliance behavior (Bulgurcu, Cavusoglu, & Benbasat, 2010; D'Arcy, Hovav, & Galletta, 2009; Li et al., 2019; Mamonov & Benbunan-Fich, 2018). Consequently, the absence of adequate security awareness exposes the organization to the risk of data breaches (Mahlaola & Van Dyk, 2016). For example, an evaluation of staff cybersecurity awareness highlighted that a common failure in data protection was walking away from organizational systems without logging off (Arain et al., 2019). Similarly, an investigation on the use of e-communication with patients in mental health care revealed limited awareness of cybersecurity risk by some clinicians (Elhai & Hall, 2015). An example was leaving an unsecured mobile phone unattended for around five minutes; the same phone was used for calling patients. Such a lack of awareness in organizations has been reported as a major cause of accidental exposure of personal information (Kwon & Johnson, 2018). Furthermore, an investigation conducted by Firdouse et al. (2018) highlighted concerns regarding safety and security in the context of using text messaging for clinical communications. They argued that regulatory guidance and increased awareness are necessary to assist physicians in determining what constitutes appropriate use of texting. Insufficient training in privacy and security awareness has been linked to a tendency to underestimate the seriousness of situations requiring protection of personal health data, consequently leading to breaches in confidentiality (Elger, 2009; Fernando & Dawson, 2009). Hence, we put forward the following proposition:

Proposition 2. Healthcare workers' lack of awareness can facilitate data breach incidents.

4.5.3. Data protection failure in third-party sources

Despite the importance of value creation by using third-party services in healthcare, there is a paucity of empirical research that analyzes the potential threats emerging from third parties as potential data breach facilitators. In our review, we found three studies that investigated this failure in the context of mHealth. An analysis of 24 mHealth apps and data sharing showed that personal health data such as blood pressure, symptoms (e.g., headache), and feelings (anxiousness) were available to third parties (Grundy et al., 2019). Such failures in data protection can facilitate data breach incidents. For instance, cybercriminals have the ability to utilize data mining and exploit the susceptibilities present in text messaging software, which frequently retains text archives for extended periods of five years or beyond (Firdouse et al., 2018). Another evaluation of mental health apps showed that some apps requested unnecessary permissions for accessing personal data.

Accepting this risky permission allows developers to read, modify, and delete data (Parker, Halter, Karliychuk, & Grundy, 2019). The above examples highlight the critical nature of informed use of third-party applications in the health context and the importance of effective management of third-party relationships from a privacy protection perspective. Hence, we put forward the following proposition:

Proposition 3. Data protection failures in third-party sources can facilitate data breach incidents.

4.5.4. Organizational characteristics

In the included studies, the main organizational characteristics related to data breaches were size, security resources, and organizational type. In healthcare settings, managerial decisions related to these factors (i.e., number of beds, and expenditures) affect the likelihood of data breaches (McLeod & Dolezel, 2018). Regarding organizational size, it is worth noting that small healthcare institutions often face constraints when allocating financial resources for investment in IT security. However, intriguingly, the evidence pertaining to data breaches in the US indicates that larger hospitals have a greater likelihood of experiencing such breaches (Gabriel et al., 2018). A lack of security resources can facilitate data breach occurrence (Kwon & Johnson, 2013a). A case in point is the study by Fernando and Dawson (2009), which revealed that privacy and security training in healthcare workplaces is unlikely to achieve functional optimization due to inadequate budgets. Kwon and Johnson (2014) demonstrated that at state and organizational levels, IT budgets were associated with proactive security investments, which in turn contributed to positive externalities (improving data protection). However, these positive impacts were diminished if the investments were coerced by government regulation pressures (Kwon & Johnson, 2014). Finally, there is a relationship between organization type and data breaches (Dolezel & McLeod, 2019). Gabriel et al. (2018) reported that the risk of data breaches is higher in pediatric and teaching hospitals compared to other hospital types. Similarly, Choi, Johnson, and Lehmann (2019) noted that major teaching hospitals are more likely to be associated with data breaches. This association can be attributed to the higher number of individuals accessing patient data; for example, due to higher clinician turnover from residency and fellowship programs, and more interactions among clinicians involved in both health delivery and medical education (Choi & Johnson, 2019; Choi et al., 2019). Hence, we put forward the following proposition:

Proposition 4. Organizational characteristics (size, security resources, and organizational type) can facilitate data breach incidents.

4.5.5. Inappropriate collective responses

In our review, we found that only a single attempt had been made to investigate a group's collective response to cybersecurity events. Borrowing the theory of social disorganization from the criminology field, Johnston et al. (2019) investigated a collective-level response to cybersecurity issues in a healthcare organization. Lack of emergent social cooperation, such as collectively identifying cybersecurity problems, can influence group performance in data protection. Effective data protection responses require collective actions, especially in healthcare where multiple data processors such as clinicians are involved in healthcare delivery. Hence, we put forward the following proposition:

Proposition 5. Inappropriate collective responses by different healthcare groups can facilitate data breach incidents.

4.6. Impacts

Healthcare organizations that experience security incidents resulting in data breaches are susceptible to a myriad of adverse consequences. Specifically, if prompt and efficient measures to rectify the breach are not implemented immediately, it can impact the attitudes and behaviors of various healthcare stakeholders, including patients, as well as hinder

the creation of business value. From our review, we found that two overarching research themes constitute the core of data breach impacts: behavioral impacts and business value impacts.

4.6.1. Behavioral impacts

Our analysis revealed four second-order themes, namely, privacy concerns expressed, health information sharing withheld, system use reduced, and trust damaged, as the behavioral impacts of data breaches.

4.6.2. Privacy concerns

The challenges associated with acquiring and managing personal health information, along with addressing privacy concerns, become increasingly apparent given the rising importance and widespread implementation of electronic records in digital healthcare (Angst & Agarwal, 2009). Patients expressed unauthorized access and hacking as a common privacy concern in the case of using EHR for service delivery (Kisekka & Giboney, 2018; Mwachofi, Khaliq, Carrillo, & Winfree, 2016). Bansal, Zahedi, and Gefen (2010) showed that when customers perceived a high level of privacy concern as a result of privacy invasion, they were reluctant to disclose information to health websites. Such behaviors inevitably pose challenges to the success of digital health. Data breaches illustrate the security vulnerabilities of health systems, which in turn lead to patients developing a negative perception of the health system's ability to safeguard their privacy. It is reasonable to assume that data breaches result in significant privacy concerns. Hence, we put forward the following proposition:

Proposition 6. Data breach incidents can result in increased patient privacy concerns.

4.6.3. Health information sharing

Patients who use digital health services have an expectation to control the privacy of their personal data. In situations where the services are perceived to be less confidential and can lead to unauthorized access, they are unlikely to disclose health information (Zhou, 2018). For instance, Schairer et al. (2019) found perceived data protection failures such as lack of transparency in data processing, fear of IT hacking, and data tracking act as mechanisms against health data sharing. Fear of failure to preserve personal health data by healthcare providers is a barrier in patients' information sharing (Agaku, Adisa, Ayo-Yusuf, & Connolly, 2014), as it may prevent organizations from improving health service delivery. A study in the context of a pandemic outbreak showed that data breach incidents constrained patient information sharing by physicians (El Emam et al., 2011). This interpretation differs from that of Walker, Johnson, Ford, and Huerta (2017), who found that the effects of privacy and security concerns on patients' personal health data sharing were minimal and constant. They explained that customers experiencing excellent care are willing to accept the potential risk of breaches associated with personal health data. This view proposes that high quality of care in some situations may reduce the perceived importance of a potential data breach. However, with an equal quality of care (*ceteris paribus*), one can expect that data breach incidents negatively affect information sharing. Hence, we put forward the following proposition:

Proposition 7. Data breach incidents can result in withholding health information sharing by patients.

4.6.4. System use

We identified a twofold relationship between system use and data breaches. Earlier, we cited examples of ineffective use as a data breach facilitator. Meanwhile, data breach incidents can also impact the future use of health systems. System use plays a pivotal role in the literature on IS and holds significant implications for the future of health systems (Recker et al., 2019; Wu & Du, 2012). It holds the potential to enhance organizational effectiveness and efficiency when users leverage the systems and their accurate information representation to accomplish

their tasks (Burton-Jones & Grange, 2013; Burton-Jones & Volkoff, 2017). However, the intended goals of the system are often unattainable in the event of system failures and privacy breaches. Poor security measures and inadequate responses to data breaches, for example, give rise to privacy concerns among users. Consequently, this can impede the effective use of digital health systems (Vrhovec & Markelj, 2018). This issue has been highlighted in the context of online patient portals within a general hospital (Tieu et al., 2015). For instance, one significant barrier to the use of portals by healthcare consumers was their concern regarding the system's vulnerability to hackers and the potential breach of confidentiality about diagnosis and medication data. Health organizations can mitigate this concern by implementing a highly secure system for disclosing health information, thereby enabling patients to use the portal more frequently and effectively. Failure to address these concerns regarding privacy breaches in organizational practices may result in reduced health portal use among patients. Hence, we put forward the following proposition:

Proposition 8. Data breach incidents can result in reduced health system use by patients.

4.6.5. Trust

While the notion of trust has been extensively explored from various angles, particularly as a factor influencing IS acceptance and use (Alalwan, Dwivedi, & Rana, 2017; Fox & Connolly, 2018; Söllner, Hoffmann, & Leimeister, 2016), further investigation is needed to understand the detrimental effects of data breaches on trust. The existing empirical evidence in digital health is currently limited. Regarding trust as an outcome of data breaches (see Fig. 2), we refer to the detrimental effects that can lead to a decline in customer trust in organizations. This decline can occur when healthcare organizations fail to effectively safeguard digital health systems and protect client data. Agaku et al. (2014) emphasized the significance of robust security measures on personal health data, as their insufficiency can diminish patients' trust and confidence in their healthcare providers. This critical issue highlights the interplay between data protection and patient trust, demanding thorough attention within the healthcare network. Emphasizing the significance and complexity of trust in the processes of patients' decision-making, Croll (2011) introduced a privacy framework and comprehensive guidelines for the healthcare sector. The objective was to establish a connection between confidentiality, trust, privacy, and security. The study's findings revealed that data breaches could potentially erode the trust that individuals placed in organizations. Furthermore, research indicated that individuals who had experienced prior data breaches tended to have less confidence in the security measures implemented by these organizations (Tieu et al., 2015). Hence, we put forward the following proposition:

Proposition 9. Data breach incidents can result in damaging stakeholder trust in a healthcare organization.

4.6.6. Business value impacts

The second theme regarding the impact of the data breach is business value. Our review revealed this impact represents two second-order themes, namely care quality, and trust repair costs. These two impacts were studied less frequently in the existing literature compared to behavioral impacts such as privacy concerns and health information sharing.

4.6.7. Care quality

The availability of data in EHR plays a vital role in integrated digital healthcare, aiding providers in maintaining and enhancing the quality of care they offer to customers (Amato, De Pietro, Esposito, & Mazzocca, 2015). Accordingly, the quality of health services may be compromised when data breaches occur or concerns about privacy and security arise. As an illustration, a study by Kisekka and Giboney (2018) revealed that privacy concerns, such as unauthorized access to EHR, have a

Table 5
Summary of the health data breaches literature.

Propositions in the model of health data breaches	Representative studies that have examined this relationship in the literature	Theories used for support in these studies
Proposition 1. Healthcare workers' noncompliant behavior can facilitate data breach incidents.	Calhoun et al. (2018), Kuo, Talley, Hung, and Chen (2017), Desouza and Valverde (2016), Sher, Talley, Cheng, and Kuo (2017), Kamoun and Nicho (2014), Desouza and Valverde (2016), Angst et al. (2017), Meehan et al. (2018)	Deterrence theory in Kuo et al. (2017) study. Swiss Cheese Model in Kamoun and Nicho (2014) study. Institutional theory in Angst et al. (2017) study. Rational choice theory in Gaia, Wang, Yoo, and ers (2020) study. The rest examined this link without an underlying theory.
Proposition 2. Healthcare workers' lack of awareness can facilitate data breach incidents.	Arain et al. (2019), Fernando and Dawson (2009), Walker, Migneault, Lindsay, and Abu-Laban (2019)	Health information system security threat lifecycle in a Fernando and Dawson (2009) study. The rest examined this link without an underlying theory.
Proposition 3. Data protection failures in third-party sources can facilitate data breach incidents.	Grundy et al. (2019), Müthing et al. (2017), Parker et al. (2019), Firdouse et al. (2018)	Examined but without an underlying theory
Proposition 4. Organizational characteristics (size, security resources, and organizational type) can facilitate data breach incidents.	Angst et al. (2017), McLeod and Dolezel (2018), Kwon and Johnson (2014), (Li, Yoo, & Kettinger, 2021), Kwon and Johnson (2013a), Bai, Jiang, and Flasher (2017a)	Institutional theory, Resource-based view, Organizational learning, Swiss cheese model, Routine activity theory
Proposition 5. Inappropriate collective responses by different healthcare groups can facilitate data breach incidents	Johnston et al. (2019)	Social Disorganization Theory
Proposition 6. Data breach incidents can result in increased patient privacy concerns.	Kisekka and Giboney (2018), Bansal et al. (2010)	Utility theory, Information systems success model
Proposition 7. Data breach incidents can result in withholding health information sharing by patients.	El Emam et al. (2011), Schairer et al. (2019)	Theory of contextual integrity, Social contract theory
Proposition 8. Data breach incidents can result in reduced health system use.	Tieu et al. (2015), Vrhovec and Markelj (2018), Choi et al. (2019), Pool, Akhlaghpour, and Fatehi (2020)	UTAUT in Tieu et al. (2015) study. Theory of Mobile Health Data Protection (MHDP) in Pool et al. (2020) study.
Proposition 9. Data breach incidents can result in damaging stakeholder trust in a healthcare organization.	Agaku et al. (2014), Croll (2011), Tieu et al. (2015)	UTAUT in Tieu et al. (2015) study. The rest examined this link without an underlying theory.
Proposition 10. Data breach incidents can result in compromising care quality.	Choi et al. (2019), Kisekka and Giboney (2018)	Examined but without an underlying theory
Proposition 11. Data breach incidents can result in incurring trust repair costs to a healthcare organization.	Choi and Johnson (2019)	Examined but without an underlying theory

detrimental impact on patients' attitudes toward EHR and their willingness to share their data. This impedes health information exchange and decreases patients' perception of care quality. Similarly, Choi et al. (2019) indicated the negative impacts of health data breaches on care quality. For example, they found an association between data breach incidents and increases in time from door to electrocardiogram in hospitals due to data breach remediation efforts. Hence, we put forward the following proposition:

Proposition 10. Data breach incidents can result in compromising care quality.

4.6.8. Trust repair costs

One of the unintended outcomes of data breaches is a violation of trust. After such breaches, organizations should take action for repairing their public reputation and brand (Gillespie & Dietz, 2009). With the introduction of data breach notification laws such as the HIPAA Breach Notification Rule and Australia's Notifiable Data Breaches (NDB) scheme, customers are becoming more aware of health data breaches. As a consequence, healthcare entities should consider the repair costs associated with failures in their data protection. A study in the US hospital context revealed that there is an association between data breaches and advertising expenditures (Choi & Johnson, 2019). This issue highlights the importance of effective data protection and remedial actions to minimize the financial burden on healthcare systems. Hence, we put forward the following proposition:

Proposition 11. Data breach incidents can result in incurring trust repair costs to a healthcare organization.

5. Discussions

With the rapid expansion of digital health technologies, the occurrence of personal health data breaches has emerged as a pressing

concern for healthcare organizations. These breaches have the potential to detrimentally affect both individual behavior and organizational business value creation. In response to this challenge, healthcare organizations are compelled by regulatory and institutional pressures to prioritize investments in security and privacy (Angst et al., 2017). Nevertheless, there persists an ongoing necessity to systematically gather evidence about the various types of breaches, the factors that enable them, and the resulting impacts of data breaches on individuals and healthcare organizations. Furthermore, there is a need to advance theoretical and contextual understanding of these phenomena of health data breaches.

We summarized our findings from a synthesis of the data-driven and theory-driven health data breach studies, in an integrative model. As illustrated in Fig. 2, this model can be used in studying data breaches and failures in health data protection. In Table 6, we summarize the paths that are implied in the literature to date as well as the theoretical support and empirical evidence. Together, Fig. 2 and Table 5 provide an integrative view of the literature conducted to date and provide a platform upon which we can develop further theory and design more integrative tests.

Our integrative model brings forth crucial insights. Our propositions 1 and 2 state that two prominent facets contributing to healthcare data breaches are noncompliance with information security policies and a lack of security awareness among healthcare workers. Proposition 1 reflects the persistent challenge of health workers' non-compliance (Gwebu et al., 2019; Lowry & Moody, 2015; Soomro et al., 2016). Habit formation, influenced by factors like self-efficacy and perceived usefulness, plays a crucial role in compliance behavior (Moody et al., 2018; Kuo et al., 2018). In practice, non-compliance can manifest through various behaviors, such as the failure to log out of accounts driven by perceptions of computational inconvenience (Eikey et al., 2015; Sarkar, Vance, Ramesh, Demestihias, & Wu, 2020). Proposition 2 complements this challenge by addressing healthcare workers' lack of

Table 6
Health data breach stakeholders.

Stakeholder	Illustrative roles
IT/security department	Chief information officer, chief information security officer, data protection officer, IT/security managers, IT/security staff
Healthcare personnel/staff	Physicians, GPs, family doctors, nurses, surgeons, allied health professionals, interns, residents, fellows, medical students, non-clinical staff (e.g., receptionists, researchers)
Healthcare manager/director	Health service managers, finance management/unit, human resource management (including education and training)
Business associate	mHealth app developers, app markets, marketing companies, medical device manufacturers, suppliers
Patient/health service customer	Chronic disease patients, patients with cancer, psychiatry outpatients, patients and their relatives

security awareness (Mahlaola & Van Dyk, 2016). Insufficient training in privacy and security awareness can lead to underestimating the need for personal health data protection, compromising confidentiality (Elger, 2009; Fernando & Dawson, 2009). These propositions collectively underscore the need for robust security policies, comprehensive awareness programs, and a deep understanding of factors influencing compliance behavior and awareness to effectively mitigate data breach risks in healthcare.

Propositions 3–5 discuss external factors contributing to healthcare data breaches. Proposition 3 highlights the risks associated with data protection failures in third-party sources, emphasizing the critical importance of informed and vigilant use of third-party healthcare applications (Grundy et al., 2019; Parker et al., 2019). Proposition 4 draws attention to the influence of organizational characteristics, such as size, security resources, and type, on data breaches (Choi et al., 2019; Gabriel et al., 2018). Larger healthcare institutions, despite their apparent robustness, are not immune to breaches. This highlights the complex interplay of organizational factors in data protection. Proposition 5 emphasizes the significance of collective responses in healthcare cybersecurity (Johnston et al., 2019). It highlights the need for seamless cooperation among stakeholders to effectively identify and mitigate cybersecurity threats. These three propositions emphasize the multifaceted nature of data breaches in healthcare and the importance of comprehensive data protection strategies that consider both external factors and internal collective responses.

Propositions 6–9 collectively address the behavioral impacts of data breaches in healthcare, shedding light on the far-reaching consequences for stakeholders. Proposition 6 focuses on heightened patient privacy concerns resulting from data breaches. The breach of trust in safeguarding sensitive health information can have significant ramifications, necessitating strategies to restore patient confidence in healthcare organizations (Bansal et al., 2010; Kisekka & Giboney, 2018). Proposition 7 discusses the impact of data breaches on health information sharing by patients, demonstrating that breaches can deter patients from disclosing health information (Agaku et al., 2014; Schairer et al., 2019). Proposition 8 explores the relationship between data breaches and reduced health system use by patients, as breaches can lead to privacy concerns (Vrhovec & Markelj, 2018). Proposition 9 delves into trust in healthcare organizations, highlighting the significance of robust security measures in preserving patient trust (Agaku et al., 2014; Tieu et al., 2015). Data breaches can erode trust, particularly when organizations fail to safeguard digital health systems and patient data effectively. Understanding the intricate relationship between data protection and trust is essential for healthcare organizations to maintain stakeholder confidence.

Finally, propositions 10 and 11 spotlight the consequences of data breaches on healthcare organizations' business value. Proposition 10 focuses on compromised care quality due to data breaches, emphasizing the importance of patient data security for maintaining high-quality healthcare (Choi & Johnson, 2019; Kisekka & Giboney, 2018). Proposition 11 discusses the financial implications of trust repair costs resulting from data breaches. Following the EU GDPR lead, regulatory frameworks in countries such as the US, Australia, and India have increased the financial penalties associated with health data breaches in

recent years (Akhlaghpour, Hassandoust, Fatehi, Burton-Jones, & Hynd, 2021). This proposition reiterated the requirement healthcare organizations must implement effective data protection measures to minimize financial costs associated with restoring trust and reputation.

These propositions collectively demonstrate the multifaceted causes and impacts of data breaches on healthcare organizations, encompassing care quality and financial considerations. Healthcare entities must proactively address these impacts to ensure quality care and manage trust repair costs, safeguarding their long-term viability and reputation.

5.1. Implications for healthcare stakeholders, practice and policy

This review study led us to identify five different stakeholders involved, contributing to, or impacted by the data breach process. The findings of our study have implications for various stakeholders involved in health data processing, from the IT/security department to health service customers. (see Table 6). More details and examples of each stakeholder, their contributions to data breach facilitators, types, and impacts are also illustrated in Appendix, Tables A3–A6.

IT/security department: IT/security departments, as stakeholders in healthcare organizations, bear the primary responsibility for preventing security incidents such as hacking attacks or unauthorized access. The IT/security department should implement effective access controls and monitoring mechanisms to mitigate noncompliant behavior among healthcare workers. In this regard, Chief Information Officers (CIOs) and Chief Information Security Officers (CISOs), Data Protection Officers (DPOs), IT/Security managers, and IT staff should prioritize fostering a culture of data security and compliance within the organization. This includes advocating for training programs where healthcare workers' noncompliant behavior and lack of awareness can facilitate data breaches. Furthermore, DPOs should lead the charge in assessing and enhancing data protection measures in third-party sources by establishing reasonable criteria for external partners and business associates. Officers in the IT/security department should ensure that organizational characteristics, such as organizational and IT department size and security resources, are taken into account when formulating information security policies. CISOs should work closely with DPOs to address data protection failures in third-party sources by establishing robust business associates' assessment and monitoring processes. Lastly, everyone, from CIOs to IT staff, should actively contribute to the development of collective response protocols to address and manage data breaches promptly.

Healthcare personnel/staff: We identified healthcare personnel, residents, and students as other stakeholders that involved or facilitated the data breach process. For instance, they contributed to data breaches through unauthorized access and lack of awareness. Health professionals, including but not limited to physicians, GPs, surgeons, and nurses, must adhere strictly to data access protocols, ensuring that only authorized personnel have access to health data. Attending regular training and compliance with information security policies are vital for the effective use of health IT. They should actively engage in cybersecurity awareness programs to stay updated on the latest data protection measures and best practices. This includes understanding the importance of strong passwords, encryption, and secure communication

channels, as well as the data protection-aware use of Artificial Intelligence (AI)-enabled medical devices. Recognizing that data breaches can compromise care quality, this stakeholder should prioritize patient data protection as an integral part of providing high-quality healthcare. This involves vigilance in protecting patient data and promptly reporting any privacy violations and suspicious security activity. Furthermore, as several actors are involved in health data processing, addressing non-compliant behavior among different actor-groups such as interns, residents, fellows, medical students, and non-clinical staff (e.g., receptionists, and researchers), is important to minimize potential data breach incidents. To achieve this, training programs should be implemented to educate them about the importance of adhering to data protection protocols and policies. For example, providing scenario-based training modules reflecting the role of these actors and conducting periodic assessments can help reinforce compliance.

Business associate: Among the external stakeholders identified were business associates such as mHealth app developers. They were involved in data protection failures as third-party sources providing tools to patients and healthcare providers. For example, they contributed to data breaches by presenting mHealth apps in mobile app markets (e.g., Google Play) without proper data protection by design which facilitated data breaches. Collaboration with business associates, such as AI app developers, EHR vendors, legal counsel, specialized cybersecurity firms, and cyber insurance providers, is crucial to addressing data protection failures stemming from third-party sources. Close monitoring and evaluation of these collaborations and partnerships can help mitigate risks and ensure compliance with data protection regulations. Regular data audits and third-party assessments can help identify data breach vulnerabilities and ensure compliance with data protection regulations such as HIPAA and GDPR.

Healthcare manager: The data breach also can have an impact on other stakeholders in healthcare such as business/finance executives and directors. The cost of data breaches can directly influence the business through data breach penalties, trust repair activities, and the need for new investments for the security of digital assets. The healthcare C-suites and the board of directors should focus on implementing comprehensive data protection strategies and allocating sufficient resources to IT and security departments. This includes ensuring proper education and training for healthcare personnel, as well as promoting a culture of responsible use of data within healthcare organizations.

Patient/health service customer: The main stakeholders in data breaches are patients. Their information can be impacted directly by data breaches, such as hacking and unauthorized access. Regarding this stakeholder group, we identified impacts such as increased privacy concerns, reduced health IT use, and damaged trust. Following a data breach incident, patients and their relatives raise privacy concerns, leading to a reluctance to share health information with healthcare providers, thereby contributing to the ineffective use of health IT for public healthcare services. To address these challenges, healthcare organizations must adopt a proactive approach. Trustworthy practices such as transparency and open communication are crucial, ensuring that patients and their relatives are informed about the remedial actions taken to rectify the breach and prevent future occurrences. Offering options for this stakeholder to enhance privacy controls, such as multi-factor authentication and end-to-end encryption communication channels, and allowing customers to have their data removed when it is no longer necessary for medical purposes, can empower patients to take an active role in protecting their health data. Additionally, educational resources and workshops focused on customer data protection can rebuild trust in the healthcare system. It is equally important to extend these efforts to patients' relatives and carers in the case of elderly and child patients, providing them with the same level of transparency and data protection reassurance. By tailoring these recommendations to patients and their relatives, healthcare organizations can work towards securing stakeholder trust, ensuring the continued use of health IT services, and reducing privacy concerns following a data breach incident.

Policy implication and policymakers: In formulating a robust policy to safeguard health data from data breaches, a collective approach must be adopted, ensuring that all stakeholders work towards the goal of data protection. This entails establishing communication channels and collaborative frameworks between IT/security departments, health managers, healthcare personnel, business associates, and health service customers. Each stakeholder's unique insights contribute to a holistic strategy, effectively mitigating potential breaches and fortifying the overall security posture. Managers and healthcare professionals should prioritize the strengthening of IT and security departments to prevent data breach incidents such as hacking attacks, theft, loss, improper disposal, and unauthorized access. Adequate IT resources, empowered by a "privacy by design" culture and training to all stakeholders, can help in effective data protection. Business associates should also hold to high standards, with contractual obligations that prioritize data privacy and security. Health managers, as leaders of healthcare organizations, must ensure that organizational resources are allocated effectively, and data protection policies are enforced rigorously. Lastly, health service customers must be engaged and informed about their rights and the measures in place to protect their data. Through this collective approach, healthcare organizations can enhance their defences against potential data breaches, creating information resilience for health data protection.

Our review highlights that healthcare personnel and staff play a significant role in data protection failures arising from unauthorized access, noncompliance, and lack of awareness. Simultaneously, the often under-resourced IT and security departments within healthcare organizations have the main responsibility of preventing security incidents and safeguarding patient data from hacking and unauthorized access. Recognizing the distinct roles of these stakeholders, policymakers and academic institutions should advocate for the development and implementation of specialized cybersecurity and privacy training programs tailored to the healthcare sector. These programs should actively engage multiple healthcare stakeholders to enhance their understanding of privacy and security protocols, reducing the risk of insider threats and cyberattacks. Our findings highlight the need for regulations and policies that promote data security and privacy in healthcare by consideration of health data breach facilitators identified in our study. These include noncompliance, lack of awareness, failures in third-party sources, inappropriate collective responses, and organizational characteristics. Collective responses to health data breaches should be emphasized, and noncompliance should be addressed through effective law enforcement mechanisms. Policymakers should also consider the unique challenges posed by emerging technologies in healthcare, such as AI, and ensure that appropriate safeguards are in place. Overall, our study highlights the importance of a multistakeholder approach to data breach management and effective compliance with data protection policies. Collaboration among stakeholders, including IT/security departments, healthcare personnel, health managers, business associates, and patients, is crucial for achieving robust data protection practices and safeguarding personal health data.

5.2. A road map for future research

Our scoping review and synthesis of the past literature highlight several directions for future research. We discuss six that we believe are particularly profitable and propose questions for future research.

Future directions 1: Multilevel analyses – Researchers can begin with our integrative model and pursue avenues for extending it. We believe a very clear way to do so would be to knit the levels of analysis that have so far been studied separately. This is clear from Table A1, which shows that researchers have examined data breach phenomena at different levels of analysis. By so doing they have shown the relevance of these phenomena at each level, but we still know very little of how the levels interrelate. Multilevel research is necessary for offering a complete picture of health data breaches. As a gap identified in the health data

breach literature, there is limited understanding of how the data breach phenomenon unfolds from a multilevel perspective (Lapointe & Rivard, 2005).

Such knowledge is crucial for managers because they need to implement coherent strategies that cut across levels. We recommend two ways that researchers could undertake this work. On one hand, they could look at the downside risk of health data breaches and theorize such breaches through the lens of ‘organizational errors’. The organizational error literature has shown that multilevel analysis can be very helpful for understanding how and why such errors occur (Goodman, 2000). On the other hand, they could adopt a positive stance and use theories of high-reliability organizations to understand how such errors can be minimized (Weick & Sutcliffe, 2007) and the multilevel process required to do so (Butler & Gray, 2006). In essence, conducting multilevel analyses in future research endeavors can offer more comprehensive explanations for why organizations are susceptible to data breaches, and provide insights into devising effective responses and mitigating risks. Possible research questions to explore:

1. How and why do data protection errors occur at individual and organizational levels and how are they related to each other?
2. How can organizations collectively minimize the risk of data breaches across a multitude of roles, from frontline health staff to IT departments and CEOs?

Future directions 2: Methods of analysis – Researchers could take a more exploratory stance and build new theories bottom-up from grounded and situated studies of health data breaches in practice. It would be advantageous to employ a mixed-method approach in this line of research. The occurrence of a data breach is a multifaceted phenomenon, and hackers employ a range of diverse and ingenious techniques to exploit vulnerabilities and target organizational systems. In order to comprehensively investigate this dynamic subject, researchers should avoid restricting themselves to any single method, but rather be prepared to use a combination of interviews, surveys, risk analysis, failure mode and effects analysis (FMEA), media discourse analysis, and observations. Possible research questions to explore:

1. What do rich qualitative (e.g., ethnography) studies tell us about the privacy protection practices enacted in health organizations?
2. How do methodological biases in sample selection affect the generalizability of data privacy research?

Future directions 3: Contributing to the IS theories – We echo the call from Lowry, Dinev, and Willison (2017) for the effective use of “powerful” IS theories, theories that “explain more than just intentions” (p. 547), in privacy and security research. As previously discussed in the results section, only 24% of the articles analyzed in our study explicitly incorporated a theory. This can be attributed to the influence of diverse disciplinary norms within the field of medical informatics. Nevertheless, we believe that directing more attention toward IS theories can result in valuable cross-disciplinary and interdisciplinary advantages (Chiasson, Reddy, Kaplan, & Davidson, 2007; Tarafdar & Davison, 2018). Specifically, upon our review, we identified several concepts and themes that were absent from theoretical frameworks or models in empirical investigations. These noteworthy omissions encompass hacking/IT incidents, loss, improper disposal, and theft as data breach types, and ineffective use and failures in third-party sources as data breach facilitators. Future research can use native IS theory such as representation theory and effective use for theorizing concepts such as hacking/IT incidents and ineffective use. This would extend these theories, because they have not been used in that way to date (see Burton-Jones and Grange, 2013; Burton-Jones, Recker, Indulska, Green, and Weber, 2017). By using these findings and delving deeper into further theorizing the impactful phenomenon of data protection in healthcare settings and other data-driven contexts, we can enhance our understanding in such a

critical area. Possible research questions to explore:

1. How can (in)appropriate cybersecurity training influence the (in) effective use of digital health systems?
2. How can bad actors (e.g., hackers, insider system abusers) take advantage of IT resource inadequacy and third-party sources to affect healthcare organizations?

Future directions 4: Stakeholder analysis – Our review revealed that the literature provides inadequate explanations of stakeholders’ roles and their contributions to data breaches and impacts (see Tables A3-A6 in the Appendix). For example, most extant empirical studies examined data breach types (e.g., unauthorized access) and data breach facilitators (e.g., lack of awareness) with a focus on healthcare professionals. They ignored providing a complete picture of different stakeholders. Neglecting to study the role of other stakeholders has portrayed a single view on data breaches. To enhance our understanding of data breaches, future studies should go beyond a single perspective and investigate multi-stakeholders including IT and information security departments, healthcare professionals, non-clinical staff, administrators, business associates, and patients. To elaborate, this integration requires explaining how different stakeholders are involved in data breaches and what the impacts of data breaches are on organizational and individual stakeholders such as the IT department, finance unit, human resource management, and patients. Possible research questions to explore:

1. How can the lack of cyber-awareness of health professionals influence IT/information security departments’ responses to security attacks (e.g., ransomware)?
2. How do diverse stakeholders (healthcare professionals, administrators, IT/security departments, patients) with different tasks and interests play a role in data breach types, facilitators, and impacts?

Future directions 5: Investigating less-explored facets of emerging themes from this review – One of the primary causes, related to the human factor, for ongoing security incidents is employees’ failure to comply with the organizational information security policy (ISP) while utilizing information resources (Liu, Wang, & Liang, 2020). However, there are major limitations in the current body of literature addressing noncompliance, particularly in health IS and the context of data breaches. In our review, existing studies have demonstrated a lack of depth in their investigation, both in terms of theoretical and practical understanding of institutional complexity and, notably, individual factors such as emotions and habits. These shortcomings highlight a knowledge gap in the health IS security and privacy domains, necessitating further research to bridge this critical gap. For instance, in a general context, there is recent evidence that emotion plays a significant role in individuals’ decision-making processes complying with security and privacy policies (Chen et al., 2021). In our review, we have not found any studies that explicitly investigate the role of emotion in health data breaches. Consequently, our understanding of this impact remains limited. Nonetheless, we can draw upon the IS research to enrich insight into the issue of noncompliance. For instance, a study conducted by Yazdanmehr, Li, and Wang (2023) sheds light on the link between emotion and noncompliance. Their investigation showed that employees experiencing stress due to ISP demands tend to resort to emotion-focused coping as a means to alleviate the stress. Consequently, this may lead to noncompliance such as violations of the ISP. Expanding on this topic of emotion and noncompliance, Xu, Luo, and Hsu (2020) examined the effects of discrete emotions such as anger and fear on employee computer-related deviant behavior. Examples of computer-related deviant behavior include attempting to hack someone’s password or access their files without permission, and altering or deleting information without their consent. They emphasized that employees who experienced anger were more prone to engaging in computer-related deviant behavior, whereas those who experienced fear were less inclined to engage in such behavior. Furthermore, concerning

emotions and noncompliance, tension exists in organizations where employees strive to comply with ISP. The tension arises from conflicting motivations: instrumental consequences, such as material benefits, versus socio-emotional consequences, such as maintaining positive social relationships. For instance, an employee may feel obligated to comply with social expectations even if it means bypassing security measures (Karjalainen, Sarker, & Siponen, 2019). This tension between socio-emotional and instrumental consequences highlights the challenges employees face in balancing the need for security and the desire to help others. The complexity of individual decision-making in the context of organizational noncompliance requires further investigation.

In addition to exploring emotion, it is also important to understand the role of habit concerning noncompliance. This aspect was not properly addressed in the reviewed health IS literature either. Few studies in the digital health context approached habit. For instance, as we discussed in the findings section, one study found that motivational aspects such as self-efficacy and perceived usefulness played a crucial role in fostering habitual behaviors, which subsequently contributed to individuals' sustained commitment to comply with the privacy policy (Kuo et al., 2018) and reduced the likelihood of noncompliance. This finding, in line with past IS research, suggests that habit can be a powerful force in shaping user behavior (Tamilmani, Rana, & Dwivedi, 2021), even when it comes to something as important as security and privacy (Moody et al., 2018). Healthcare organizations need to understand how IS use habits are formed and how such habits influence noncompliance. This understanding will enable them to develop strategies aimed at helping individuals develop more compliant habits. Staff noncompliance behavior is not always driven by conscious decision-making; sometimes, individuals unknowingly fall into patterns of behavior. Therefore, in this behavioral process, motivational factors can influence the development of habitual behavior. This is an important research direction that can help us understand the mechanism of habitual behavior by altering individuals' motivations that facilitate security and privacy policy compliance.

We propose that future research explore questions such as:

1. How does the interplay between motivational factors and habit formation influence healthcare staff's noncompliance with security and privacy policies?
2. How do individuals navigate the paradoxical tension between habit formation and conscious decision-making when it comes to compliance with security and privacy policies in the digital health context?
3. How do conflicting emotions (e.g., fear and anger) impact healthcare EHR users' behavior regarding compliance with data protection regulations?
4. What are the underlying mechanisms through which habit formation impacts noncompliance behavior among healthcare professionals, and how can these mechanisms be altered to promote greater adherence to security and privacy policies?
5. How does the tension between the material benefits of IS security behaviors and socio-emotional consequences (maintaining positive social relationships) influence employees' compliance with ISP and their likelihood of engaging in noncompliant behavior, specifically concerning unauthorized access to or disclosure of health information?

One of the primary contributors to data breaches frequently arises from an organization's inability to adhere to established data protection standards. The studies in our review mainly focused on employee noncompliance from an individual-level perspective. While employee noncompliance with organizational policies is always a risk factor, organizations' noncompliance with network and data protection measures enhances their susceptibility to malicious attacks. Hence, an organization's noncompliance can play a critical role in facilitating potential data breaches. Organizations must prioritize and implement comprehensive compliance strategies to minimize the risk of data breaches and enhance

their overall security posture. While this noncompliance is widely recognized, fewer insights from the health IS literature warrant further research. Thus, we propose that future research can explore questions such as:

1. What are the institutional factors that contribute to organizational noncompliance with data protection measures, and how can regulators and healthcare organization stakeholders collectively address these factors?
2. How do compliance strategies impact an organization's resilience to prevent and respond to data breaches?

Despite our review covering studies conducted during the COVID-19 pandemic from 2019 to 2022, there remains a significant gap in the literature regarding the investigation of the health data breach process, particularly the crucial concept of compliance in this public health emergency context. It is imperative to understand the influence of public health emergencies, such as COVID-19, on compliance, and its implications for data protection. This knowledge is vital for effectively addressing and mitigating potential breaches in the healthcare sector. The COVID-19 pandemic has had a significant impact on health organizations' service delivery and supply chain capabilities, especially in terms of remote patient-clinician interaction via telehealth and the increased reliance on digital technologies. However, this shift towards remote work and digital transformation, along with the widespread use of IS, has also brought about heightened security risks and vulnerabilities (Dwivedi et al., 2020; Wong, Lee, Tan, Ooi, & Sohal, 2022). One notable consequence of the pandemic is the introduction of remote work policies by many organizations, which are expected to continue post pandemic. In this context, employee compliance with organizational ISP becomes crucial in mitigating cybersecurity threats. The implementation of work-from-home policies has the potential to bring about shifts in information security culture and employee compliance with ISP (Butler & Brown, 2023). Furthermore, sudden spikes in demand, due to major shocks such as the COVID-19 pandemic, can have ripple effects on other organizational processes, including security metrics (Boh, Constantinides, Padmanabhan, & Viswanathan, 2023). In their efforts to ensure continuity and timeliness of service delivery during major shocks, organizations must also ensure that their security measures are not compromised. It is crucial to understand how these environmental disruptions and major shocks such as the COVID-19 pandemic affect cultural aspects and compliance behavior. Additionally, exploring the role of patients as stakeholders would further contribute to our understanding of the complexities and implications of health data breaches during public health emergencies. For instance, the shift towards remote healthcare services due to the pandemic has led to sensitive patient data being transmitted over the internet, making patients susceptible to phishing, impersonation attacks, and other cybersecurity threats. In this context, the role of the patient as a stakeholder becomes increasingly significant. However, the lack of awareness and noncompliance by patients can become major vulnerabilities. As remote care delivery may continue to be a prominent approach in the future, it is crucial to consider and incorporate these factors into our understanding of health data breaches.

We propose that future research can explore questions such as:

1. How do public health emergencies such as the COVID-19 pandemic influence noncompliance with data protection regulations in the healthcare sector?
2. What are the potential long-term implications of the COVID-19 pandemic on data protection regulations and compliance in the healthcare sector?
3. What institutional mechanisms and contextual factors contribute to noncompliance with organizational ISP among healthcare employees during the implementation of remote work policies in response to the COVID-19 pandemic?

4. How might the increased use of telehealth during the COVID-19 pandemic shape future regulatory frameworks and guidelines related to data protection and compliance in the healthcare sector?
5. How can digital technologies such as AI support institutional policies and governance structures to effectively mitigate noncompliance behaviors during significant disruptions like the COVID-19 pandemic? Furthermore, how can this be achieved while addressing the diverse needs and interests of the healthcare stakeholders involved?
6. How can effective strategies be developed to increase patient awareness and promote best practices of data protection during public health emergencies, particularly in the context of remote healthcare services, and how would these strategies help mitigate the risk of health data breaches?

Future directions 6: Boundary-breaching opportunities – Finally, we advocate several opportunities for future research that crosses current boundaries on health data breaches. This boundary-breaching opportunity includes studying new concepts and structuring specific relations, emerging digital health contexts, and possible feedback loops.

Our review provides a useful overview of what themes and concepts have been studied in health data breaches. To pursue research into this domain we need to understand new concepts such as lack of data protection by design and its relation with ineffective use. Also, understanding the roles of these facilitators in data breach types such as hacking and unauthorized access can be a key opportunity for future researchers. As a starting point, researchers can borrow insights from IS theories, such as the effective use of IT, to contribute to the stream of knowledge on IS security and IS design.

Furthermore, emerging digital health contexts such as the Internet of Things (IoT), AI, Large language models (LLMs), blockchain, big data, the metaverse, and telehealth are inadequately examined in past studies on health data breaches. This limitation of knowledge could affect our understanding of real-world phenomena. For example, the ubiquitous use of telehealth due to COVID-19 and health data maximization through the use of IoT, AI, and big data can increase data breach risks (Cohen & Mello, 2019; Gia, Nawaz, Querata, Tenhunen, & Westerlund, 2019; Jalali, Landman, & Gordon, 2021). We believe these digital health contexts deserve attention for future investigations. To ensure that this risk in the context is well understood, researchers should contextualize their studies where the impacts of data breaches are significant for both patients and organizations. For example, by exploring a given data breach type/facilitator (e.g., hacking, ineffective use, or unauthorized access) in a particular IT context (e.g., IoT, telehealth, or AI), researchers can advance the knowledge within a boundary-breaking landscape. Also, Application of blockchain-enabled technologies in healthcare offers promising solutions for addressing privacy and security concerns and mitigating health data breaches (Yaqoob, Salah, Jayaraman, & Al-Hammadi, 2022; Zhang, White, Schmidt, Lenz, & Rosenbloom, 2018). By integrating blockchain in health data systems health data can be securely stored and accessed only by authorized parties, reducing the risk of data breaches such as unauthorized access. However, particularly with the emergence of medical deepfakes (Solaiyappan & Wen, 2022), future research can investigate the practical implications and challenges of implementing blockchain to minimize health data breaches. Future research can investigate the key technical, and institutional considerations associated with the integration of blockchain technology in healthcare data management, and how these can be effectively used for data sharing while ensuring compliance with existing healthcare data protection regulations.

In light of the growing concerns surrounding privacy, responsible AI, and the potential dark side of AI (Dennehy et al., 2023; Kumar, Dwivedi, & Anand, 2021; Rana, Chatterjee, Dwivedi, & Akter, 2022), several key areas warrant attention for future research. It is crucial to deepen our collective understanding of AI in healthcare and its responsible use to address the dark side of this emerging technology. The dark sides, and

risks associated with AI, including privacy invasion and information leakage, necessitate the implementation of responsible AI in healthcare (Kumar et al., 2021). Moreover, recent reports of privacy violations, biases, and inaccurate data used to train AI algorithms have raised concerns regarding the ethical and responsible use of AI (Dennehy et al., 2023). For example, there are questions about the limits of the 'right to be forgotten' as identified by modern privacy regulations such as GDPR (Akhlaghpour et al., 2021), if personal data is already used for AI training. Therefore, future research should delve into the development of ethical, transparent, and accountable AI solutions that prioritize patient data privacy and build trust. This involves exploring how to align AI models with stakeholder expectations, using relevant and quality data, practicing data minimization, and building information resilience for better data protection. These practices are essential considerations in minimizing data breach risks, building trust, and ensuring responsible AI in healthcare. Also, understanding the limitations of generative AI, such as ChatGPT, as discussed in the literature (Dwivedi et al., 2023; Haupt & Marks, 2023; Paul, Ueno, & Dennis, 2023), is crucial to mitigate reputation and legal risks, as well as for protecting health data privacy. For example, there can be heightened health data protection risks if LLM models are used for preparing more persuasive phishing messages to vulnerable patients, or generative AI tools are used for inserting fake tumors in medical images. Hence, in order to address and mitigate potential negative consequences, an important and critical direction for future research is investigating the responsible use of emerging AI systems, such as ChatGPT, in healthcare while considering the dark side of cybersecurity and privacy risks.

When researchers contextualize data breaches, they should also consider feedback loops. Extant literature has studied the impacts associated with data breaches, such as health information sharing and trust. However, the links and feedback loops from data breach impact to data breach types or facilitators are missing. Future research could identify, for instance, how effectively managing one data breach impact can minimize the probability of future data breaches. In addressing this missing line of research, future research needs to examine the development of data protection practices and capabilities (e.g., effective use, proactive response) as feedback-based learning from failures in the past. Possible research questions to explore:

1. How can lack of data protection by design facilitate data breaches and impede care quality?
2. How can ineffective use of IoT and AI increase the likelihood of data breaches in healthcare?
3. How can organizations learn from failures in data protection and develop capabilities that enable effective responses to data breaches?
4. How can AI models in healthcare be aligned with stakeholder expectations and ethical considerations to ensure responsible use and minimize the risks of health data breaches?
5. How can information resilience practices be incorporated into the development of responsible AI solutions in healthcare to protect patient data privacy and mitigate potential breaches?
6. How can healthcare stakeholders understand and effectively address the privacy risks associated with the use of generative AI (e.g., ChatGPT) and ensure the responsible use of AI?

5.3. Limitations

We acknowledge the limitations of our study. Firstly, in this review, we considered personal health data breaches from a socio-technical perspective. However, other perspectives such as purely technical or legal perspectives also play important roles in managing data breaches. A data breach is a complex phenomenon, and issues such as encryption and breach detection algorithms, mathematical models and also legal regimes and ethical considerations are relevant. They need to be studied in separate articles. Secondly, we did not include articles published before 2009. As explained earlier, aside from practical issues, we believe

the year 2009 is a reasonable choice for a starting point because the digitalization of health records and EHR/EMR diffusion accelerated around this time, partly due to the influential HITECH Act in the US, and similar initiatives around the globe. Thirdly, a contextual review has inherent limitations. While our contextualized model offers valuable insights into data breaches within the specific healthcare context, we acknowledge the importance of incorporating general knowledge of data breaches. For instance, existing literature on data breaches in general, such as the review included in the study by Goode et al. (2017), provides a wealth of relevant information. By integrating our contextualized findings with this existing body of knowledge, we can develop a more comprehensive model that strengthens our understanding and helps prevent the loss of essential general knowledge. To address this inherent limitation of our study, future research can merge contextualized knowledge from this review with general knowledge beyond healthcare to develop a more comprehensive understanding of data breaches in diverse contexts. Finally, our review may be limited in terms of generalizability due to strict inclusion and exclusion criteria regarding the health data breach process. One limitation of the paper is the potential omission of valuable insights related to the broad research topic of privacy and security. While we established specific inclusion and exclusion criteria to ensure the relevance of the selected papers to the research question, this rigorous approach might have resulted in the exclusion of studies that could have provided interesting findings from other disciplines and contexts. By excluding studies that were not directly relevant to the process of health data breaches, the context of healthcare, systems, and users, we may have missed valuable perspectives, alternative approaches, or innovative ideas that could have enriched the review article. This limitation implies that the current findings may not fully encompass the breadth of the literature on data breaches in general. Future research could employ text mining techniques to generate keyword clouds or graphs, enabling the exploration of potentially interesting insights. By conducting such an analysis, potential avenues for future research could identify hidden yet relevant information that would extend and increase the impact of the review paper.

6. Concluding remarks

This paper conducts a thorough investigation, spanning 2009–2022,

into empirical and conceptual research about health data breaches. Employing a thematic analysis and integrative approach, we have discerned an extensive range of themes encompassing personal health data breaches, as well as enabling factors and unintended outcomes. Thus, personal health data breaches represent a covert menace to the successful implementation of meaningful digital health transformations. We anticipate that an exploration of the various categories, enabling factors, and consequences of health data breaches can aid healthcare organizations in formulating proactive measures and implementing remedial actions, thereby fortifying their information resilience in the event of a health data breach.

A famous quote among cybersecurity experts is “a data breach alone is not a disaster, but mishandling it is”.² The objective of our paper is to provide a more comprehensive understanding of the various types, facilitators, and impacts of health data breaches. We encourage researchers to consider our findings and concentrate on the less-explored yet significant concepts of data breaches when theorizing in extreme contexts of healthcare. Furthermore, we posit that security practitioners can employ our model as a valuable resource to mitigate the risks associated with breaches of personal health data within their respective organizations. By diligently overseeing the types and enablers of data breaches identified in our analysis, they can gain a comprehensive understanding and effectively prepare for these potential risks.

CRediT authorship contribution statement

JP: Conceptualization, Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing, Data extraction, Synthesis and Interpretation. **SA:** Conceptualization, Formal analysis, Methodology, Supervision, Writing – review & editing, Synthesis and Interpretation. **FF:** Conceptualization, Formal analysis, Methodology, Supervision, Writing – review & editing. **ABJ:** Conceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare no competing financial interests or personal relationships that could have influenced the paper. There is no conflict of interest that could affect their objectivity in conducting this review.

Appendix A

Table A1

Characteristics of reviewed articles.

No.	Author/s	Country	Setting	System	Method	Level of analysis	Theory
1	Almulihi et al. (2022)	US	Multiple setting	Health information system (HIS)	Quantitative	Organizational	N/A
2	Belfrage, Helgesson, and Lynøe (2022)	Sweden	Multiple setting	Electronic health record (EHR)	Quantitative	Individual	N/A
3	Gioulekas et al. (2022)	Greece, Portugal, and Romania	Multiple setting	HIS	Quantitative	Individual	N/A
4	Heath, Porter, and Silvera (2022)	US	Hospital	HIS	Quantitative	Organizational	N/A
5	Horvath, ucci, and James (2022)	UK	Public Health	Mobile health (mHealth)	Quantitative	Individual	N/A
6	Kandasamy, Srinivas, Achuthan, and Rangan (2022)	N/A or Not mentioned	Multiple setting	HIS	Quantitative	Organizational	N/A

(continued on next page)

² The original quote is attributed to Serene Davis of Beazley Group (<http://cyberrescue.co.uk/library/quotes>).

Table A1 (continued)

No.	Author/s	Country	Setting	System	Method	Level of analysis	Theory
7	Jamal et al. (2022)	Saudi Arabia	Multiple setting (related to clinical photography)	Multiple systems (mHealth, social media, hospital cameras system)	Quantitative	Individual	N/A
8	Kim, Park, Park, Chun, and Wiles (2022)	South Korea	Hospital	HIS	Quantitative	Individual	Prosocial rule-breaking perspective
9	Lee (2022)	US	Multiple setting	HIS	Quantitative	Organizational	N/A
10	Marchang and Di Nuovo (2022)	N/A	Aged Care	Robotic System	Review	N/A	N/A
11	Owusu (2022)	Ghana	Other (informal care)	mHealth	Mixed-Methods	Individual	N/A
12	Park, Kim, and Wiles (2022)	US	Education	HIS	Quantitative	Individual	Deterrence theory, and health information security awareness (HISA)
13	Yeo and Banfield (2022)	US	Multiple setting	HIS	Qualitative	Organizational	N/A
14	Alzahrani (2021)	India	Hospital	Online platform (including web apps)	Quantitative	Individual	N/A
15	Bressler, Grudnikoff, Bressler, Tamez, and Zampella (2021)	US	Dermatology	Social media	Quantitative	Individual	N/A
16	Choi and Johnson (2021)	US	Hospital	HIS	Quantitative	Organizational	N/A
17	Churi, Pawar, and Abdulmuhsin (2021)	India	Multiple setting	HIS	Quantitative	Individual	N/A
18	Dang, Guo, Guo, Wang, and Xie (2021)	China	N/A	mHealth	Quantitative	Individual	Privacy calculus theory, social exchange theory, social support theory
19	Elhoseny et al. (2021)	N/A	N/A	Multiple systems (e.g., patient monitoring, and implantable devices)	Review	N/A	N/A
20	Giannouchos et al. (2021)	US	Other (health research)	Privacy-enhancing software	Qualitative	Individual	Privacy paradox
21	Idris, Kassim, Ridzuan, and Shamsulbaharin (2021)	Malaysia	Education	Social media	Quantitative	Individual	N/A
22	Joyce, Roman, Miller, Jeffries, and Miller (2021)	N/A	Oncology practices	HIS	Review	N/A	N/A
23	Li et al. (2021)	US	Hospital	HIS	Quantitative	Organizational	Routine activity theory
24	Looten et al. (2021)	France	Hospital	Email	Quantitative	Individual	NA
25	Patel and Makaryus (2021)	N/A	Cardiology	Multiple systems (e.g., implantable c)	Review	N/A	N/A
26	Thomasian and Adashi (2021)	N/A	N/A	Multiple systems (e.g., wearables devices)	Review	N/A	N/A
27	Ahmed, Jagsi, Gutheil, and Katz (2020)	Multiple (e.g. Australia, Canada, India)	Multiple setting	Social media	Mixed-Methods	Individual	N/A
28	Choi, Johnson, and Lee (2020)	US	Hospital	HIS	Quantitative	Organizational	N/A
29	Du and Wang (2020)	China	Genetic Test Services	N/A	Review	Policy	N/A
30	Gaia et al. (2020)	US	Education	HIS	Quantitative	Individual	Rational choice theory
31	Hathaliya and Tanwar (2020)	N/A	N/A	Multiple systems (e.g., IoT, telehealth, cloud, and EHR)	Review	N/A	N/A
32	Jung, Lee, Kim, and Lee (2020)	South Korea	Public health (pandemic)	Contact tracing	Quantitative	Individual	N/A
33	Offner, Sitnikova, Joiner, and MacIntyre (2020)	Australia	Other (healthcare organizations)	HIS	Review	N/A	N/A
34	Parmar, Patond, Rathod, and Ninave (2020)	India	Education	HIS	Quantitative	Individual	N/A
35	Ploug (2020)	N/A	Other (research)	N/A	Review	N/A	N/A
36	Pool et al. (2020)	N/A	Multiple setting	mHealth	Review	N/A	Mobile Health Data Protection (MHDP)
37	Sarkar et al. (2020)	US	Hospital	EHR	Mixed-Methods	Individual	Deterrence Theory
38	Seh et al. (2020)	US	Multiple setting	HIS	Quantitative	Organizational	N/A
39	Beltran-Aroca, Labella, Font-Ugalde, and Girela-Lopez (2019)	Spain	Hospital	HIS	Quantitative	Individual	N/A
40	Huang, Chen, Tang, and Huang (2019)	Taiwan	N/A	mHealth	Quantitative	Individual	Expectation-confirmation theory
41	Kim and Kwon (2019)	US	Hospital	EHR	Quantitative	Organizational	Human factor theory and routine activity theory
42	Yaqoob, Abbas, and Atiquzzaman (2019)	N/A	N/A	Multiple systems (e.g., implantable devices, wearables, and on-site equipment)	Review	N/A	N/A

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Table A1 (continued)

No.	Author/s	Country	Setting	System	Method	Level of analysis	Theory
43	Walker et al. (2019)	Canada	Emergency Medicine	mHealth	Quantitative	Individual	N/A
44	Schairer et al. (2019)	US	Multiple-setting	Multiple systems (e.g., Genetic testing, Biobanking)	Qualitative	Individual	Theory of contextual integrity
45	Chernyshev et al. (2019)	N/A	N/A	EMR	Review	N/A	N/A
46	Rieger et al. (2019)	US	Psychiatry Outpatients	mHealth	Quantitative	Individual	N/A
47	Parker et al. (2019)	N/A	Emergency Medicine	mHealth	Other (apps content analysis)	Privacy	N/A
48	Koczkodaj et al. (2019a)	US	Multiple setting (covered by HIPAA)	HIS	Quantitative	Organizational	N/A
49	Koczkodaj et al. (2019b)	US	Multiple setting (covered by HIPAA)	EHR	Quantitative	Organizational	N/A
50	Johnston et al. (2019)	US	Other (Healthcare research organization)	HIS	Qualitative	Group	Social Disorganization Theory
51	Hattingh, King, Hope, and George (2019)	Australia	Pharmacy	Multiple systems (fax, Facebook)	Quantitative	Individual	N/A
52	Grundty et al. (2019)	N/A	N/A	mHealth	Other (app Traffic, content, and network analysis)	IT artefact	N/A
53	Dolezel and McLeod (2019)	US	Multiple setting (covered by HIPAA)	HIS	Quantitative	Organizational	N/A
54	Choi et al. (2019)	US	Hospital	HIS	Quantitative	Organizational	N/A
55	Choi and Johnson (2019)	US	Hospital	HIS	Quantitative	Organizational	N/A
56	Arain et al. (2019)	Canada	Other (healthcare organization)	HIS	Quantitative	Individual	N/A
57	Ronquillo, Erik Winterholler, Cwikla, Szymanski, and Levy (2018)	US	Multiple setting (covered by HIPAA)	HIS	Quantitative	Organizational	N/A
58	Stark, Gewald, Lautenbacher, Haase, and Ruff (2018)	Germany	Medical center	HIS	Quantitative	Individual	N/A
59	Vrhovec and Markelj (2018)	Slovenia	Hospital	mHealth	Quantitative	Individual	N/A
60	Jiang and Bai (2018)	US	Multiple setting	HIS	Quantitative	Organizational	N/A
61	McCoy and Perlis (2018)	US	Multiple setting	HIS	Quantitative	Organizational	N/A
62	Coventry and Branley (2018)	N/A	N/A	N/A	Review	N/A	N/A
63	Kuziemy, Gogia, Househ, Petersen, and Basu (2018)	N/A	N/A	Telehealth	Review	N/A	N/A
64	McLeod and Dolezel (2018)	US	Hospital	HIS	Quantitative	Organizational	Swiss cheese model
65	Kuo et al. (2018)	Taiwan	Medical center	EMR	Quantitative	Individual	Self-determination theory and habit-related perspective
66	Calhoun et al. (2018)	US	Education	N/A	Other	Individual	N/A
67	Kisekka and Giboney (2018)	US	Other	EMR	Quantitative	Individual	Information systems success
68	Meehan et al. (2018)	US	Education	EMR	Mix	Individual	N/A
69	Firdouse et al. (2018)	Canada	Hospital	mHealth	Quantitative	Individual	N/A
70	Yaraghi and Gopal (2018)	US	Multiple setting (Business associates of the health care providers)	HIS	Quantitative	Organizational	N/A
71	Gabriel et al. (2018)	US	Hospital	HIS	Quantitative	Organizational	N/A
72	Guo et al. (2018)	Canada	Education	mHealth	Quantitative	Individual	N/A
73	Zhou (2018)	China	Other	Online platform	Quantitative	Individual	N/A
74	Hepp et al. (2018)	Canada	Multiple-setting	HIS	Qualitative	Organizational	N/A
75	Parks, Xu, Chu, and Lowry (2017)	US	Multiple setting (hospitals and other healthcare organizations)	HIS	Qualitative	Organizational	The Unintended consequences of privacy safeguard enactment
76	Tomossy, Bending, and Maluga (2017)	Australia	Public health	HIS	Review	N/A	N/A
77	Angst et al. (2017)	US	Hospital	HIS	Quantitative	Organizational	Institutional theory
78	Kuo et al. (2017)	Taiwan	Hospital	EMR	Quantitative	Individual	Deterrence theory (criminal domain)
79	Bai, Jiang, and Flasher (2017b)	US	Hospital	HIS	Quantitative	Organizational	N/A
80	Wu et al. (2017)	N/A	Mental health	mHealth	Review	N/A	N/A
81	Zandona and Thompson (2017)	US	Hospital	N/A	Review	N/A	N/A
82	Kruse et al. (2017)	N/A	N/A	N/A	Review	N/A	N/A

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Table A1 (continued)

No.	Author/s	Country	Setting	System	Method	Level of analysis	Theory
83	Müthing et al. (2017)	Germany, France and the United Kingdom	N/A	mHealth	Other	IT artefact	N/A
84	Hassidim et al. (2017)	Israel	Medical center	EMR	Quantitative	Individual	N/A
85	Drolet et al. (2017)	US	Other	mHealth	Quantitative	Individual	N/A
86	Moggridge (2017)	UK	N/A	Ultrasound	Other	IT artefact	N/A
87	Sher et al. (2017)	Taiwan	Hospital	EMR	Quantitative	Individual	Integrative: Health belief model (HBM), preventive health behavior
88	Walker et al. (2017)	US	N/A	EHR	Quantitative	Individual	N/A
89	Martorell, do Nascimento, and Garrafa (2016)	Brazil	N/A	Social media	Qualitative	Individual	N/A
90	Blanke and McGrady (2016)	US	N/A	Portable devices	Quantitative	Organizational	N/A
91	Zeadally, Isaac, and Baig (2016)	N/A	N/A	N/A	Review	N/A	N/A
92	Perakslis and Stanley (2016)	US	Multiple-setting	HIS	Quantitative	Organizational	N/A
93	Luna, Rhine, Myhra, Sullivan, and Kruse (2016)	N/A	N/A	N/A	Review	N/A	N/A
94	Mwachofi et al. (2016)	US	Physicians' offices	EHR	Qualitative	Individual	N/A
95	Mahlaola and Van Dyk (2016)	South Africa	Hospital	PACS	Quantitative	Individual	N/A
96	Desouza and Valverde (2016)	Canada	Multiple-setting	N/A	Qualitative	Organizational	N/A
97	Elhai and Hall (2015)	US	Mental health	mHealth	Quantitative	Individual	N/A
98	Liu, Musen, and Chou (2015)	US	Multiple-setting	HIS	Quantitative	Organizational	N/A
99	Tieu et al. (2015)	US	Hospital	Online platform	Quantitative	Individual	Integrative: Theory of acceptance and use of technology
100	Prochaska et al. (2015)	US	Education	mHealth	Quantitative	Individual	N/A
101	Eikey et al. (2015)	US	Hospital	EHR	Mixed	Individual	N/A
102	Kwon and Johnson (2014)	US	Other (healthcare organization)	HIS	Quantitative	Organizational	Organizational learning
103	Wikina (2014)	US	Multiple-setting	HIS	Quantitative	Organizational	N/A
104	Sethi, Lane, Newton, Egan, and Ghosh (2014)	UK	Hospital	Desktop computer	Qualitative	Individual	N/A
105	Agaku et al. (2014)	US	N/A	EMR	Quantitative	Individual	N/A
106	Kamoun and Nicho (2014)	N/A	N/A	N/A	Review	N/A	Swiss Cheese Model (SCM)
107	Kwon and Johnson (2013a)	US	Hospital	HIS	Quantitative	Organizational	Resource-based view
108	Baskaran et al. (2013)	UK	Hospital	HIS	Mixed	Organizational	N/A
109	Kapoor and Nazareth (2013)	US	Multiple-setting	HIS	Quantitative	Organizational	N/A
110	Renaud (2012)	UK	Other (NHS)	HIS	Quantitative	Individual	N/A
111	Gupta and Sharman (2012)	US	Multiple-setting	HIS	Quantitative	Organizational	N/A
112	Albarrak (2012)	Saudi Arabia	Hospital	HIS	Quantitative	Individual	N/A
113	Liginlal, Sim, Khansa, and Paul (2012)	US	Multiple setting	HIS	Qualitative	Organizational	Norman's action theory
114	Xu, Dinev, Smith, and Hart (2011)	US	Education	Online platform (website)	Quantitative	Individual	Communication privacy management theory
115	El Emam et al. (2011)	Canada	Family Medicine	N/A	Mixed	Individual	Social contract theory
116	Croll (2011)	Australia	Multiple-setting	HIS	Other	Organizational	N/A
117	Bansal et al. (2010)	US	Education (University)	Online platform (website)	Quantitative	Individual	Integrative: Utility theory, trust theory
118	Elger (2009)	Switzerland	Other	N/A	Quantitative	Individual	N/A
119	Fernando and Dawson (2009)	Australia	Hospital	HIS	Mixed	Individual	Health information system security threat lifecycle (HISSTL)
120	(Simon, Evans, Benjamin, Delano, & Bates, 2009)	US	Multiple setting	HIS	Qualitative	Individual	N/A

Table A2

Types, facilitators, and impacts of data breaches.

Aggregate dimensions	Themes	Examples of studies using this theme
Data breach types	Unauthorized access/disclosure	Marchang and Di Nuovo (2022), Belfrage et al. (2022), Almulihi et al. (2022), Heath et al. (2022), Horvath et al. (2022), Jamal et al. (2022), Kim et al. (2022), Lee (2022), Owusu (2022), Park et al. (2022), Yeo and Banfield (2022), Alzahrani (2021), Bressler et al. (2021), Choi and Johnson (2021), Dang et al. (2021), Giannouchos et al. (2021), Idris et al. (2021), Looten et al. (2021), Thomasian and Adashi (2021), Ahmed et al. (2020), Koczkodaj et al. (2019a), Dolezel and McLeod (2019), Chernyshev et al. (2019), Hattingh et al. (2019), Koczkodaj et al. (2019b), Grundy et al. (2019), Calhoun et al. (2018), Jiang and Bai (2018), McCoy and Perlis (2018), Firdouse et al. (2018), Yaraghi and Gopal (2018), Gabriel et al. (2018), Guo et al. (2018), Wu et al. (2017), Zandona and Thompson (2017), Blanke and McGrady (2016), Zeadally et al. (2016), Mahlaola and Van Dyk (2016), Desouza and Valverde (2016), Perakslis and Stanley (2016), Liu et al. (2015), Tieu et al. (2015), Prochaska et al. (2015), Eikey et al. (2015), Wikina (2014), Kamoun and Nicho (2014), Gupta and Sharman (2012), Baskaran et al. (2013), Kapoor and Nazareth (2013), Elger (2009)
	Hacking/IT incident	Marchang and Di Nuovo (2022), Gioulekas et al. (2022), Almulihi et al. (2022), Kandasamy et al. (2022), Heath et al. (2022), Lee (2022), Yeo and Banfield (2022), Choi and Johnson (2021), Elhoseny et al. (2021), Joyce et al. (2021), Li et al. (2021), Thomasian and Adashi (2021), Koczkodaj et al. (2019a), Chernyshev et al. (2019), Dolezel and McLeod (2019), Koczkodaj et al. (2019b), McCoy and Perlis (2018), Yaraghi and Gopal (2018), Jiang and Bai (2018), Coventry and Branley (2018), Kuziemsky et al. (2018), Gabriel et al. (2018), Zandona and Thompson (2017), Kruse et al. (2017), Zeadally et al. (2016), Perakslis and Stanley (2016), Luna et al. (2016), Elhai and Hall (2015), Liu et al. (2015), Gupta and Sharman (2012), Kamoun and Nicho (2014), Wikina (2014), Kapoor and Nazareth (2013)
	Improper disposal	Heath et al. (2022), Lee (2022), Koczkodaj et al. (2019a), Koczkodaj et al. (2019b), Dolezel and McLeod (2019), Jiang and Bai (2018), McCoy and Perlis (2018), Yaraghi and Gopal (2018), Gabriel et al. (2018), Zandona and Thompson (2017), Moggridge (2017), Perakslis and Stanley (2016), Liu et al. (2015), Wikina (2014), Kamoun and Nicho (2014), Kapoor and Nazareth (2013)
	Loss	Heath et al. (2022), Lee (2022), Koczkodaj et al. (2019a), Koczkodaj et al. (2019b), Dolezel and McLeod (2019), Yaraghi and Gopal (2018), Jiang and Bai (2018), McCoy and Perlis (2018), Gabriel et al. (2018), Zandona and Thompson (2017), Blanke and McGrady (2016), Perakslis and Stanley (2016), Elhai and Hall (2015), Liu et al. (2015), Wikina (2014), Kamoun and Nicho (2014), Kapoor and Nazareth (2013)
	Theft	Heath et al. (2022), Lee (2022), Yeo and Banfield (2022), Koczkodaj et al. (2019a), Chernyshev et al. (2019), Koczkodaj et al. (2019b), Dolezel and McLeod (2019), Yaraghi and Gopal (2018), McCoy and Perlis (2018), Jiang and Bai (2018), Gabriel et al. (2018), Zandona and Thompson (2017), Perakslis and Stanley (2016), Blanke and McGrady (2016), Elhai and Hall (2015), Liu et al. (2015), Gupta and Sharman (2012), Kamoun and Nicho (2014), Wikina (2014), Kapoor and Nazareth (2013)
Data breach facilitators	Noncompliance	Kuo et al. (2018), Calhoun et al. (2018), Hepp et al. (2018), Kuo et al. (2017), Sher et al. (2017), Desouza and Valverde (2016), Eikey et al. (2015), Sethi et al. (2014), Kamoun and Nicho (2014), Renaud (2012), Albarrak (2012), Arain et al. (2019), Meehan et al. (2018), Angst et al. (2017), Desouza and Valverde (2016), Kamoun and Nicho (2014)
	Lack of awareness	Kandasamy et al. (2022), Gioulekas et al. (2022), Churi et al. (2021), Walker et al. (2019), Arain et al. (2019), Firdouse et al. (2018), Hepp et al. (2018), Mahlaola and Van Dyk (2016), Desouza and Valverde (2016), Elhai and Hall (2015), Kamoun and Nicho (2014), Elger (2009), Fernando and Dawson (2009)
	Third-party sources	Grundy et al. (2019), Parker et al. (2019), Firdouse et al. (2018), Muthing et al. (2017)
	Inappropriate collective responses	Johnston et al. (2019)
Behavioral impact	Organizational characteristics	Heath et al. (2022), Li et al. (2021), Choi et al. (2020), Dolezel and McLeod (2019), Choi et al. (2019), McLeod and Dolezel (2018), Bai et al. (2017b), Kwon and Johnson (2013a), Kwon and Johnson (2014), Fernando and Dawson (2009), Gabriel et al. (2018), Kamoun and Nicho (2014)
	Privacy concerns	Kisekka and Giboney (2018), Mwachofi et al. (2016), El Emam et al. (2011), Bansal et al. (2010)
	Health information sharing	Schairer et al. (2019), Rieger et al. (2019), Kisekka and Giboney (2018), Zhou (2018), Coventry and Branley (2018), Walker et al. (2017), Agaku et al. (2014), El Emam et al. (2011), Bansal et al. (2010)
Business value impact	System use	Choi et al. (2019), Vrhovec and Markelj (2018), Tieu et al. (2015)
	Trust	Belfrage et al. (2022), Coventry and Branley (2018), Tieu et al. (2015), Agaku et al. (2014), Croll (2011)
	Care quality	Choi et al. (2019), Kisekka and Giboney (2018)
	Trust repair costs	Choi and Johnson (2019)

Table A3

Stakeholders for data breach types.

Stakeholder	Data breach types				
	Hacking or IT incident	Unauthorized access/disclosure	Loss	Theft	Improper disposal
IT/security department	Kandasamy et al. (2022), Almulihi et al. (2022), Gioulekas et al. (2022), Heath et al. (2022), Lee (2022), Marchang and Di Nuovo (2022), Koczkodaj et al. (2019a), Koczkodaj et al. (2019b), Dolezel and McLeod (2019), McCoy and Perlis (2018), Yaraghi and Gopal (2018), Jiang and Bai (2018), Liu et al. (2015), Wikina (2014), Kapoor and Nazareth (2013), Choi et al. (2019), Gabriel et al. (2018), Gupta and Sharman (2012)	Heath et al. (2022), Lee (2022), Marchang and Di Nuovo (2022), Gabriel et al. (2018), Gupta and Sharman (2012), Koczkodaj et al. (2019a), Koczkodaj et al. (2019b), Dolezel and McLeod (2019), McCoy and Perlis (2018), Yaraghi and Gopal (2018), Jiang and Bai (2018), Liu et al. (2015), Wikina (2014), Kapoor and Nazareth (2013), Choi et al. (2019), Baskaran et al. (2013)	Heath et al. (2022), Lee (2022), Gabriel et al. (2018), Koczkodaj et al. (2019a), Koczkodaj et al. (2019b), Dolezel and McLeod (2019), McCoy and Perlis (2018), Yaraghi and Gopal (2018), Jiang and Bai (2018), Liu et al. (2015), Wikina (2014), Kapoor and Nazareth (2013), Choi et al. (2019), Blanke and McGrady (2016)	Heath et al. (2022), Lee (2022), Koczkodaj et al. (2019a), Koczkodaj et al. (2019b), Dolezel and McLeod (2019), McCoy and Perlis (2018), Yaraghi and Gopal (2018), Jiang and Bai (2018), Liu et al. (2015), Wikina (2014), Kapoor and Nazareth (2013), Choi et al. (2019), Gabriel et al. (2018), Blanke and McGrady (2016), Gupta and Sharman (2012)	Heath et al. (2022), Lee (2022), Gabriel et al. (2018), Koczkodaj et al. (2019a), Koczkodaj et al. (2019b), Dolezel and McLeod (2019), McCoy and Perlis (2018), Yaraghi and Gopal (2018), Jiang and Bai (2018), Liu et al. (2015), Wikina (2014), Kapoor and Nazareth (2013), Choi et al. (2019), Blanke and McGrady (2016)
Healthcare personnel/staff	Belfrage et al. (2022), Almulihi et al. (2022),	Jamal et al. (2022), Kim et al. (2022), Owusu (2022),	Elhai and Hall (2015), Blanke and McGrady	Blanke and McGrady (2016), Koczkodaj et al.	Elhai and Hall (2015); Koczkodaj et al. (2019a),

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Table A3 (continued)

Stakeholder	Data breach types				
	Hacking or IT incident	Unauthorized access/ disclosure	Loss	Theft	Improper disposal
Health manager Business associate	Gabriel et al. (2018), Koczkodaj et al. (2019a), Koczkodaj et al. (2019b), Dolezel and McLeod (2019), McCoy and Perlis (2018), Yaraghi and Gopal (2018), Jiang and Bai (2018), Liu et al. (2015), Wikina (2014), Kapoor and Nazareth (2013), Gupta and Sharman (2012), Choi et al. (2019)	Prochaska et al. (2015), Tieu et al. (2015), Mahlaola and Van Dyk (2016), Calhoun et al. (2018); Firdouse et al. (2018); Guo et al. (2018), Hattingh et al. (2019), Koczkodaj et al. (2019a), Koczkodaj et al. (2019b), Dolezel and McLeod (2019), McCoy and Perlis (2018), Yaraghi and Gopal (2018), Jiang and Bai (2018), Liu et al. (2015), Wikina (2014), Kapoor and Nazareth (2013), Choi et al. (2019), Elger (2009), Baskaran et al. (2013), Grundy et al. (2019), Muthing et al. (2017)	(2016), Koczkodaj et al. (2019a), Gabriel et al. (2018), Koczkodaj et al. (2019b), Dolezel and McLeod (2019), McCoy and Perlis (2018), Yaraghi and Gopal (2018), Jiang and Bai (2018), Liu et al. (2015), Wikina (2014), Kapoor and Nazareth (2013), Choi et al. (2019)	(2019a), Koczkodaj et al. (2019b), Gabriel et al. (2018), Dolezel and McLeod (2019), McCoy and Perlis (2018), Yaraghi and Gopal (2018), Jiang and Bai (2018), Liu et al. (2015), Wikina (2014), Kapoor and Nazareth (2013), Choi et al. (2019), Elhai and Hall (2015), Gupta and Sharman (2012)	Koczkodaj et al. (2019b), Dolezel and McLeod (2019), McCoy and Perlis (2018), Yaraghi and Gopal (2018), Jiang and Bai (2018), Liu et al. (2015), Wikina (2014), Kapoor and Nazareth (2013), Blanke and McGrady (2016), Choi et al. (2019), Gabriel et al. (2018)
					Moggridge (2017)

Table A4

Stakeholders for data breach facilitators.

Stakeholder	Data breach facilitators				
	Noncompliance	Lack of awareness	Third-party	Inappropriate collective responses	Organizational characteristics
IT/security department	Desouza and Valverde (2016); Sher et al. (2017)	Desouza and Valverde (2016)			Heath et al. (2022), Choi et al. (2019), Kwon and Johnson (2014), McLeod and Dolezel (2018), Gabriel et al. (2018), Bai et al. (2017b), Kwon and Johnson (2013a)
Healthcare personnel/staff	Kuo et al. (2018), Hepp et al. (2018), Desouza and Valverde (2016), Eikey et al. (2015), Sethi et al. (2014), Albarrak (2012), Renaud (2012), Arain et al. (2019), Meehan et al. (2018), Desouza and Valverde (2016)	Gioulekas et al. (2022), Hepp et al. (2018), Walker et al. (2019), Arain et al. (2019), Firdouse et al. (2018), Desouza and Valverde (2016), Mahlaola and Van Dyk (2016), Elhai and Hall (2015), Elger (2009), Fernando and Dawson (2009)	Firdouse et al. (2018)	Johnston et al. (2019)	Heath et al. (2022), Bai et al. (2017b), Choi et al. (2019), McLeod and Dolezel (2018), Gabriel et al. (2018), Fernando and Dawson (2009)
Health manager Business associate	Kuo et al. (2018), Hepp et al. (2018), Desouza and Valverde (2016)	Hepp et al. (2018), Desouza and Valverde (2016)	Parker et al. (2019), Grundy et al. (2019), Muthing et al. (2017)	Johnston et al. (2019)	

Table A5

Stakeholders for data breach impacts.

Stakeholder	Data breach impacts					
	Privacy concerns	System use	Trust	Health information sharing	Care quality	Trust repair costs
IT/security department			Croll (2011)		Choi et al. (2019)	
Healthcare personnel/staff	El Emam et al. (2011)	Vrhovec and Markelj (2018)	Belfrage et al. (2022), Croll (2011)	El Emam et al. (2011)	Choi et al. (2019)	
Health manager					Choi et al. (2019)	Choi and Johnson (2019)
Patients/health service customers	Kisekka and Giboney (2018), Mwachofi et al. (2016), Bansal et al. (2010)	Tieu et al. (2015)	Belfrage et al. (2022), Agaku et al. (2014); Tieu et al. (2015)	Schairer et al. (2019), Rieger et al. (2019), Kisekka and Giboney (2018), Zhou (2018), Agaku et al. (2014), Bansal et al. (2010)	Choi et al. (2019); Kisekka and Giboney (2018)	

Table A6

A detailed stakeholder analysis of data breaches.

Stakeholder	Level	Type	Facilitator	Impact	Representative study
Physicians and residents	Individual	X	Lack of awareness		Walker et al. (2019)
Patients	Individual	X	X	Health information sharing	Schairer et al. (2019)
Psychiatry outpatients	Individual	X	X	Health information sharing	Rieger et al. (2019)
mHealth app developers	Policy	X	Third-party	X	Parker et al. (2019)
IT/security department, healthcare personnel/staff (general)	Organizational	Hacking or IT incident, improper disposal, loss, theft, unauthorized access/disclosure	X	X	Koczkodaj et al. (2019a)
Community pharmacists and pharmacy interns	Individual	Unauthorized access/disclosure	X	X	Hattingh et al. (2019)
mHealth app developers, parent companies, Google Play store, marketing companies	IT artefact	Unauthorized access/disclosure	Third-party	X	Grundy et al. (2019)
Hospital IT/security department, hospital personnel/staff, admins, patient with acute myocardial infarction	Organizational	Hacking or IT incident, improper disposal, loss, theft, unauthorized access/disclosure	Organizational characteristics	System use, Care quality	Choi et al. (2019)
Hospital finance department	Organizational	X	X	Trust repair costs	Choi and Johnson (2019)
IT/security department, healthcare personnel/staff (general)	Organizational	Hacking or IT incident, improper disposal, loss, theft, unauthorized access/disclosure	X	X	Koczkodaj et al. (2019b)
IT/security department, healthcare personnel/staff (general)	Organizational	Hacking or IT incident, improper disposal, loss, theft, unauthorized access/disclosure	X	X	Dolezel and McLeod (2019)
IT/security department, healthcare personnel/staff (general)	Organizational	Hacking or IT incident, improper disposal, loss, theft, unauthorized access/disclosure	X	X	McCoy and Perlis (2018)
IT/security department, healthcare personnel/staff (general)	Organizational	Hacking or IT incident, improper disposal, loss, theft, unauthorized access/disclosure	X	X	Yaraghi and Gopal (2018)
IT/security department, healthcare personnel/staff (general)	Organizational	Hacking or IT incident, improper disposal, loss, theft, unauthorized access/disclosure	X	X	Jiang and Bai (2018)
IT/security department, healthcare personnel/staff (general)	Organizational	Hacking or IT incident, improper disposal, loss, theft, unauthorized access/disclosure	X	X	Liu et al. (2015)
IT/security department, healthcare personnel/staff (general)	Organizational	Hacking or IT incident, improper disposal, loss, theft, unauthorized access/disclosure	X	X	Wikina (2014)
IT/security department, healthcare personnel/staff (general)	Organizational	Hacking or IT incident, improper disposal, loss, theft, unauthorized access/disclosure	X	X	Kapoor and Nazareth (2013)
Healthcare researchers, business development unit, admins	Group	X	Inappropriate collective responses	X	Johnston et al. (2019)
Hospital finance department	Organizational	X	X	Trust repair costs	Choi and Johnson (2019)
IT/security department, physicians	Organizational	X	Organizational characteristics	X	McLeod and Dolezel (2018)
Healthcare personnel/staff (including Medical Doctor, nursing staff, clerical staff, and non-regulated healthcare professionals)	Individual	X		X	Arain et al. (2019)
Hospital personnel (e.g., physicians, nurses)	Individual	X	X	System use	Vrhovec and Markelj (2018)
Healthcare professionals and admin staff	Individual	X	Noncompliance	X	Kuo et al. (2018)
Physician assistant students	Individual	Unauthorized access/disclosure	X	X	Calhoun et al. (2018)
Patients with cancer	Individual	X	X	Privacy concerns, health information sharing, care quality	Kisekka and Giboney (2018)
First-year podiatric residents	Individual	X		X	Meehan et al. (2018)
Staff surgeons (vascular, plastics, urology, and general surgery)	Individual	Unauthorized access/disclosure	Lack of awareness, third-party	X	Firdouse et al. (2018)
Hospital (e.g., pediatric, teaching, larger hospitals) IT/security department, healthcare personnel/staff	Organizational	Hacking or IT incident, improper disposal, loss, theft, unauthorized access/disclosure	Organizational characteristics	X	Gabriel et al. (2018)
Medical students, residents, and fellows	Individual	Unauthorized access/disclosure	X	X	Guo et al. (2018)
Online cancer community (patients or their relatives)	Individual	X	X	Health information sharing	Zhou (2018)
Clinical and nonclinical staff (e.g., physicians, nurses), management staff, admins, educators	Individual	Unauthorized access/disclosure	Lack of awareness, noncompliance	X	Hepp et al. (2018)

(continued on next page)

Table A6 (continued)

Stakeholder	Level	Type	Facilitator	Impact	Representative study
Hospital IT/security department, hospital personnel/staff	Organizational	X		X	Angst et al. (2017)
Hospital nurses	Individual	X	Noncompliance	X	Kuo et al. (2017)
mHealth app developers	IT artefact	Unauthorized access/disclosure	Third-party	X	Müthing et al. (2017)
Medical and paramedical personnel (students, nurses, consultant (senior) physician, interns, residents)	Individual	Unauthorized access/disclosure	X	X	Hassidim et al. (2017)
Surgeons	Individual	Unauthorized access/disclosure	X	X	Drolet et al. (2017)
Ultrasound manufacturers, suppliers	IT artefact	Improper disposal	X	X	Moggridge (2017)
Hospital information management staff	Individual	X	Noncompliance	X	Sher et al. (2017)
Patients	Individual	X	X	Health information sharing	Walker et al. (2017)
Hospital IT/security department, hospital personnel/staff	Organizational	X	Organizational characteristics	X	Bai et al. (2017b)
IT/security department, training and education department	Organizational	Unauthorized access/disclosure	Lack of awareness, noncompliance	X	Desouza and Valverde (2016)
Hospital health professionals (doctors, radiologists, radiographers, student radiographers, nurses, paramedics), radiology department	Individual	Unauthorized access/disclosure	Lack of awareness	X	Mahlaola and Van Dyk (2016)
Patients	Individual	X	X	Privacy concerns	Mwachofi et al. (2016)
IT/security department, healthcare personnel/staff (general)	Organizational	Loss, theft, unauthorized access/disclosure	X	X	Blanke and McGrady (2016)
Clinical psychologists (practitioners, researchers)	Individual	Loss, theft, unauthorized access/disclosure	Lack of awareness	X	Elhai and Hall (2015)
Chronic disease patients, caregivers	Individual	Unauthorized access/disclosure	X	System use, trust	Tieu et al. (2015)
Internal medicine residents	Individual	Unauthorized access/disclosure	X	X	Prochaska et al. (2015)
Hospital IT staff, EHR users (physician, resident/intern, care coordinator, nurses, pharmacists, social workers, therapists)	Individual	Unauthorized access/disclosure	Noncompliance	X	Eikey et al. (2015)
IT/security department	Organizational	X	Organizational characteristics	X	Kwon and Johnson (2014)
IT/security department	Organizational	X	Organizational characteristics	X	Kwon and Johnson (2013a)
Hospital clinical staff	Individual	X	Noncompliance	X	Sethi et al. (2014)
Patients	Individual		X	Trust, health information sharing	Agaku et al. (2014)
Maternity services managers, IT governance manager, associate director, Midwifery managers, and staff	Organizational	Unauthorized access/disclosure	X	X	Baskaran et al. (2013)
Nurses (academic hospital)	Individual	X	Noncompliance	X	Albarak (2012)
IT/security department, healthcare personnel/staff (general)	Organizational	Hacking or IT incident, theft, unauthorized access/disclosure	X	X	Gupta and Sharman (2012)
Healthcare personnel/staff (general)	Individual	X	Noncompliance	X	Renaud (2012)
Family doctors	Individual	X	X	Privacy concerns, health information sharing	El Emam et al. (2011)
IT/security department, healthcare personnel/staff (general)	Organizational	X	X	Trust	Croll (2011)
Patients	Individual	X	X	Privacy concerns, health information sharing	Bansal et al. (2010)
Physicians	Individual	Unauthorized access/disclosure	Lack of awareness	X	Elger (2009)
Clinicians (allied health, nurses, doctors) in teaching hospitals	Individual	X	Lack of awareness, organizational characteristics	X	Fernando and Dawson (2009)

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